

# Weak Error Analysis and Small-Noise Optimization of Stochastic Linear Multistep Methods (SLMMs) for Itô and Lévy SDEs

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## Abstract

We study the weak convergence of stochastic linear multistep methods (SLMMs) for solving stochastic differential equations (SDEs). While SLMMs have been previously studied under the lens of strong convergence, their weak convergence properties remain largely unexplored. We develop a general framework for computing the weak error expansion of arbitrary SLMMs using the Komolgorov Backward Partial Differential Equation and techniques from the theory of linear recurrences. This framework enables systematic analysis of both existing and newly proposed SLMMs. In particular, we construct SLMMs that achieve improved weak accuracy in small noise regimes. We extend our analysis to the broader class of Lévy-driven SDEs and derive precise weak error bounds for a new class of multistep schemes based on interpolation of drift and diffusion terms. Numerical simulations confirm theoretical predictions and illustrate regimes in which multistep methods significantly outperform standard one-step approaches in weak approximation accuracy.

## Summary

Many real-world systems, such as stock prices and weather patterns, are influenced by randomness. Hence, their behavior is impossible to predict exactly. To forecast these systems, we use computer simulations based on mathematical models called *stochastic differential equations*. These simulations take small steps forward in time, using known information to estimate future behavior. One-step methods only use knowledge about the current state of the system for their prediction. This research focuses on the more advanced “multistep methods,” which also take into account past behavior. We develop a new mathematical framework to analyze the accuracy of these multistep methods, especially in cases where randomness plays a small but important role. Using this framework, we proved that several key methods are not only reliable but can outperform one-step approaches. We also develop and analyze our own multistep method that is more accurate than previously known methods. These results allow scientists to model uncertainty and better predict complex, unpredictable systems.

# 1 Introduction

Stochastic differential equations (SDEs) are fundamental tools for modeling systems influenced by randomness. They arise naturally in physics, biology, quantitative finance, and other domains where uncertainty plays a critical role. A typical one-dimensional Itô SDE is of the form

$$dX_t = f(X_t, t) dt + g(X_t, t) dW_t,$$

where  $f$  and  $g$  are the drift and diffusion coefficients, respectively, and  $W_t$  is a standard Brownian motion. Since closed-form solutions are rare, such equations are typically studied through numerical approximation.

Classical one-step methods for numerical approximation, such as Euler–Maruyama and Milstein, are widely used and well understood. Their strong and weak convergence properties are documented extensively; see, for example, [1, Chapter 5] and [2, Chapter 14]. However, one-step methods rely only on information at the current timestep, potentially limiting their accuracy and stability.

By contrast, multistep methods incorporate information from several previous timesteps and are central to the numerical solution of deterministic ordinary differential equations (ODEs) due to their improved efficiency and accuracy. The stochastic analogues—stochastic linear multistep methods (SLMMs)—offer a promising yet underexplored alternative in the stochastic setting. While SLMMs can improve drift approximation without requiring higher-order stochastic integrals, challenges with stability and convergence have limited their widespread adoption. Prior work has primarily focused on strong convergence for specific two-step SLMMs under restrictive assumptions [3], with comparatively little attention given to weak convergence.

Weak approximation is of particular importance in applications where expectations of functionals of the solution are of interest, such as option pricing in finance or moment estimation in stochastic models of population dynamics [4]. This motivates a systematic study of the weak behavior of SLMMs.

Moreover, many real-world systems experience not just continuous fluctuations but also sudden jumps, which are better captured by SDEs driven by Lévy processes (processes like the Wiener process but with discontinuous jumps). The analysis of weak convergence for such jump-driven systems is more delicate. Existing results on Lévy-driven SDEs focus largely on one-step schemes; see [5], [6], and [7] for surveys and algorithmic developments, and [8], [9] for results on strong approximation.

This paper develops a general framework to analyze the weak error of SLMMs, both in the Brownian and Lévy-driven settings. The approach is based on using the Kolmogorov backward equation to turn quantities based on SDEs into quantities based on parabolic partial differential equations (PDEs) and yields a systematic algorithm for computing the full weak error expansion of a broad class of SLMMs. We apply this framework to existing two-step methods, derive new mul-

timestep schemes optimized for small noise regimes, and extend the analysis to Lévy-driven systems. Numerical simulations are consistent with the theoretical results and demonstrate regimes in which multistep schemes significantly outperform one-step methods in terms of weak error.

We review background on weak approximation and SLMMs in Section 2. In Section 3, we introduce the general algorithm for computing weak error expansions. We apply the theory to specific schemes, including Adams–Bashforth variants, and derive their weak order behavior in Section 4, showing that, in the small-noise case, SLMMs provide asymptotic improvement over their one-step analogs. In Section 5, we extend the framework to Lévy-driven SDEs. Proofs of the longer results are deferred to the appendices; short proofs appear in the main text.

## 2 Preliminaries

In order to approximate solutions to an SDE, we define a discrete temporal grid on  $I = [0, T]$ , with constant stepsize  $h$ , of the form  $0 = t_0 < t_1 < t_2 < \dots < t_N = T$ . We define  $h$  to be  $T/N$  so that  $t_\ell = \ell \cdot h$  for  $\ell = 0, 1, \dots, N$ . We make discrete approximations  $X_{t_0}, X_{t_1}, \dots, X_{t_N}$ . For brevity, we denote  $X_i := X_{t_i}$  for indices  $i$ . A  $k$ -step SLMM takes the form (for  $\ell \geq k$ )

$$\sum_{j=0}^k \alpha_j X_{\ell-j} = h \sum_{j=1}^k \beta_j f(X_{\ell-j}, t_{\ell-j}) + \sum_{j=1}^k \Gamma_j(X_{\ell-j}, t_{\ell-j}) I^{t_{\ell-j}, t_{\ell-j+1}}, \quad (1)$$

where the  $\alpha_j$  and  $\beta_j$  are constants. We set  $\alpha_0 = 1$ . The starting values  $X_1, \dots, X_{k-1}$  are produced by a one-step starter method; see assumption (A4) below. Every diffusion term  $\Gamma_j(x, t) I^{t, t+h}$  is a finite sum of terms each containing an appropriate function  $q$  of  $x$  and  $t$  multiplied by a multiple Wiener integral over  $[t, t+h]$ , i.e., it takes the general form

$$\Gamma_j(x, t) I^{t, t+h} = q_j(x, t) I_r^{t, t+h} + q_j(x, t) I_{r_1, r_2}^{t, t+h} + \dots$$

A general multiple Wiener integral is given by

$$I_{r_1, r_2, \dots, r_j}^{t, t+h}(y) = \int_t^{t+h} \int_t^{s_1} \dots \int_t^{s_{j-1}} y(X_{s_j}, s_j) dW_{r_1}(s_j) \dots dW_{r_j}(s_1),$$

where  $r_i \in \{0, 1, \dots, m\}$  and  $dW_0(s) = ds$ . If  $y \equiv 1$  we omit  $y$  and write  $I_{r_1, r_2, \dots, r_j}^{t, t+h}$ . Note that the integral  $I_r^{t, t+h}$  is simply the increment  $W_r(t+h) - W_r(t)$  of the scalar Wiener process  $W_r$ . The term  $I^{t, t+h}$  denotes the collection of multiple Wiener integrals associated with the interval  $[t, t+h]$ .

We introduce two common SLMMs defined in [3] that will be studied later in the paper.

**Example 2.1.** The two-step Milstein-Adams-Bashforth method (MAB2) is given by the relation

$$X_\ell - X_{\ell-1} = h \left( \frac{3}{2} f(X_{\ell-1}, t_{\ell-1}) - \frac{1}{2} f(X_{\ell-2}, t_{\ell-2}) \right) + g(X_{\ell-1}, t_{\ell-1}) I_1^{t_{\ell-1}, t_\ell} \\ + (g \partial_x g)(X_{\ell-1}, t_{\ell-1}) I_{1,1}^{t_{\ell-1}, t_\ell}.$$

**Example 2.2.** Two-step Maruyama schemes take the general form

$$\sum_{j=0}^2 \alpha_j X_{\ell-j} = h \sum_{j=1}^2 \beta_j f(X_{\ell-j}, t_{\ell-j}) + \sum_{j=1}^2 \gamma_j g(X_{\ell-j}, t_{\ell-j}) I_1^{t_{\ell-j}, t_{\ell-j+1}}$$

for coefficients  $\alpha_i, \beta_i, \gamma_i$ .

Let us denote the mean square norm of a random variable as  $\|Z\|_{L_2} = (\mathbb{E}[|Z|^2])^{1/2}$ . A scheme is mean square convergent with order  $\gamma$  if

$$\max_{\ell=1,2,\dots,N} \|X(t_\ell) - X_\ell\|_{L_2} \leq Ch^\gamma$$

for some constant  $C$  independent of our partition of  $I$ , where  $X(t_\ell)$  denotes the true solution to the SDE at  $t = t_\ell$  and  $X_\ell$  is our approximation.

Buckwar and Winkler show in [3] that two-step Maruyama schemes are mean-square convergent with order  $\frac{1}{2}$ , assuming certain consistency conditions on the  $\alpha_i, \beta_i, \gamma_i$  (and that even higher accuracy can be obtained when the noise term is small). This is done by analyzing the local error accumulated at each step, then computing global error based on local errors.

The main quantity we study is the *signed weak error*

$$\mathcal{E}(T, h) = \mathbb{E}[\phi(X_N)] - \mathbb{E}[\phi(X(T))]$$

for a smooth function  $\phi : \mathbb{R} \rightarrow \mathbb{R}$ , together with its magnitude  $\text{Err}(T, h) = |\mathcal{E}(T, h)|$ . The scheme has weak order  $\gamma$  if  $\text{Err}(T, h) \leq C_\phi h^\gamma$  for a constant  $C_\phi$  depending only on  $\phi, T$ , and the coefficients. We call a *weak error expansion* an equation of the form

$$\mathcal{E}(T, h) = c_0 h^\gamma + c_1 h^{\gamma+1} + \dots + c_r h^{\gamma+r} + O(h^{\gamma+r+1}),$$

where the constants  $c_i$  depend on the function  $\phi$  used. (The expansion must be stated for the signed error, since the coefficients  $c_i$  carry signs.) The weak error expansion of one-step schemes is understood. Talay and Tubaro [10] compute the weak error expansion of classical one-step schemes (such as Euler-Maruyama), and Iguchi and Yamada [11] compute the weak error expansion of a more involved scheme. Due to the Feynman-Kac formula [12] the main idea is to use the solution

$u(t, x)$  to the Kolmogorov Backward Partial Differential Equation (KB-PDE) [10]:

$$\begin{cases} \partial_t u + \mathcal{L}u = 0, \\ u(T, x) = \phi(x). \end{cases}$$

Here,  $\mathcal{L}$  is the infinitesimal generator of the SDE. The solution to this PDE is

$$u(t, x) = \mathbb{E}[\phi(X(T)) \mid X_t = x].$$

Therefore,

$$\mathcal{E}(T, h) = \mathbb{E}[u(T, X_N)] - \mathbb{E}[u(0, X_0)] = \sum_{i=1}^N \mathbb{E}[u(t_i, X_i) - u(t_{i-1}, X_{i-1})].$$

In other words, to understand the error in approximation of our scheme, we simply need to understand the behavior of  $u$  under small perturbations dependent on the scheme. We understand the one-step case; we can bound  $u(t_\ell + h, X_\ell + (X_{\ell+1} - X_\ell))$  using a two-variable Taylor expansion or other related methods (for example using integration and the Itô-Taylor expansion). The difficulty with multistep methods is that the differences  $X_{\ell+1} - X_\ell$  are harder to track, because they may not be directly expressed using terms dependent on  $f$  and  $g$ . We overcome this difficulty using linear recurrence theory.

We require consistency and stability conditions on the schemes in order for them to attain a weak error expansion. The stability condition is the *Dahlquist root condition*, introduced by Dahlquist in [13] for Ordinary Differential Equations (ODEs). Buckwar and Winkler deduced in [3] that this condition must hold for multistep approximation schemes in order for them to attain a certain mean square convergence order. In the convention of (1) (where  $\alpha_0$  multiplies the newest iterate), the *characteristic polynomial* of the scheme is

$$\rho(\zeta) = \sum_{j=0}^k \alpha_j \zeta^{k-j} = \zeta^k + \alpha_1 \zeta^{k-1} + \cdots + \alpha_k.$$

The SLMM (1) satisfies the Dahlquist root condition if all roots of  $\rho$  lie on or within the unit circle, and the roots on the unit circle are simple. We say the scheme satisfies the *strict root condition* if, in addition,  $\zeta = 1$  is the only root on the unit circle. The root condition ensures that a sequence  $a_n$  satisfying the homogeneous recurrence  $a_n = -\sum_{j=1}^k \alpha_j a_{n-j}$  remains bounded; the strict root condition ensures, further, that the memory kernels introduced in Section 3 decay geometrically.

**Standing assumptions.** The following are assumed throughout, and are invoked as (A1)–(A4); results requiring more state so explicitly.

(A1) (*Consistency*.)  $\sum_{j=0}^k \alpha_j = 0$ , i.e.  $\rho(1) = 0$ , and the strict root condition holds.

(A2) (*Explicitness.*)  $\beta_0 = 0$  and  $\Gamma_0 \equiv 0$ , as already reflected in (1).

(A3) (*Regularity.*)  $f, g \in C_b^m(\mathbb{R} \times I)$ , and  $\phi \in C^m(\mathbb{R})$  with polynomially bounded derivatives, for an order  $m$  specified by each result. Under these conditions the KB-PDE solution  $u$  lies in  $C^{1,m}$  with polynomially bounded spatial derivatives, and the scheme has uniformly bounded moments of all orders on  $[0, T]$ ; see [2, Chapter 4] and [10].

(A4) (*Starter.*)  $X_1, \dots, X_{k-1}$  are produced by a one-step method with local weak error  $O(h^2)$  and uniformly bounded moments (a single Euler–Maruyama step suffices).

Condition (A1) is necessary for the recurrence reduction of Section 3 to exist at all. Condition (A3) replaces an incorrect claim in an earlier version of this paper that no smoothness of  $f, g$  is required: every Taylor step below uses smoothness of  $u$ , which is inherited from smoothness of  $f, g$ , and  $\phi$ .

We additionally study Lévy-driven stochastic differential equations. The general form of a Lévy-driven SDE is

$$dX_t = h(X_{t-}) dZ_t,$$

where  $h : \mathbb{R} \rightarrow \mathbb{R}$  is a measurable function and  $Z_t$  is a real-valued Lévy process, which may include both continuous and jump components. By the Lévy–Itô decomposition and the Lévy–Khintchine characterization [14], we can equivalently express such SDEs in the more explicit form:

$$dX_t = f(X_{t-}) dt + g(X_{t-}) dW_t + \int_{|z| < 1} \gamma(X_{t-}, z) \tilde{J}(dt, dz) + \int_{|z| \geq 1} \delta(X_{t-}, z) J(dt, dz),$$

where:

- $f : \mathbb{R} \rightarrow \mathbb{R}$  and  $g : \mathbb{R} \rightarrow \mathbb{R}$  are the drift and diffusion coefficients,
- $\tilde{J}(dt, dz) = J(dt, dz) - \nu(dz) dt$  is the compensated Poisson random measure for small jumps  $|z| < 1$ ,
- $\gamma : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$  is the small-jump coefficient,
- $\delta : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$  is the large-jump coefficient, and
- $\nu(dz)$  is the Lévy measure, which governs the intensity and distribution of the jumps.

This representation separates the contributions of continuous and discontinuous components and enables precise analysis and numerical approximation.

One may define a  $k$ -step SLMM for Lévy-driven SDEs in the same way as was done for continuous SDEs. With the same grid on  $I$ , the general form is

$$\sum_{j=0}^k \alpha_j X_{\ell-j} = h \sum_{j=1}^k \beta_j f(X_{\ell-j}, t_{\ell-j}) + \sum_{j=1}^k \Gamma_j(X_{\ell-j}, t_{\ell-j}) I^{t_{\ell-j}, t_{\ell-j+1}} + \sum_{j=1}^k \Xi_j(X_{\ell-j}, t_{\ell-j}).$$

Here the  $\alpha_i, \beta_i$ , and  $\Gamma_i$  are as before. The functions  $\Xi_j$  are any functions meant to approximate the jump terms (we do not specify a general form).

**Example 2.3.** The Euler-Maruyama scheme for Lévy Driven SDEs, a 1-step SLMM, is given by

$$\begin{aligned} X_{n+1} = & X_n + hf(X_n, t_n) + \Delta W_{n+1} g(X_n, t_n) \\ & + \int_{|z|<1} \gamma(X_n, z) \tilde{J}((t_n, t_{n+1}], dz) + \int_{|z|\geq 1} \delta(X_n, z) J((t_n, t_{n+1}], dz), \end{aligned}$$

where  $\Delta W_{n+1} = W(t_{n+1}) - W(t_n)$ .

A common example of a nonlinear choice for  $\Xi_j$  is that made by the Asmussen-Rosiński approximation [15], in which the small jumps are approximated by Brownian motion.

The weak error expansion of Lévy-driven SDEs can be understood in the same sense as the continuous case. The KB-PDE is still well-defined (with  $\mathcal{L}$  now an integro-differential operator), and one can again turn the question of weak error into a question about PDEs.

### 3 Algorithm for Weak Error Estimation

To obtain a weak error expansion for an SLMM, we must first estimate the local deviations  $X_{i+1} - X_i$ . Denote  $\Delta x_\ell = X_\ell - X_{\ell-1}$ , and write  $\mathbb{E}_n[\cdot] := \mathbb{E}[\cdot | \mathcal{F}_{t_n}]$  for the conditional expectation given the filtration at time  $t_n$ . We say that  $X \stackrel{\mathbb{E}}{=} Y$  for two random variables  $X, Y$  if  $\mathbb{E}_{n-1}[X] = \mathbb{E}_{n-1}[Y]$ , where the conditioning time  $t_{n-1}$  is the left endpoint of the step under consideration. We can then Taylor expand

$$\begin{aligned} u(t+h, x + \Delta x_\ell) - u(t, x) & \stackrel{\mathbb{E}}{=} \sum_w \frac{\Delta w}{|w|!} \partial_w u \\ & \stackrel{\mathbb{E}}{=} \sum_{a+b \geq 1} \frac{1}{a! b!} h^a (\Delta x_\ell)^b \partial_t^a \partial_x^b u. \end{aligned} \tag{2}$$

Here, the summation in the first line is over all words  $w$  taken from the alphabet  $\mathcal{A} = \{t, x\}$ . The quantity  $\Delta w$  is the product of all  $\Delta c$  over characters  $c$  in  $w$ . One can define  $\partial_w u$  in a similar fashion. Notice that the partial derivatives commute by smoothness assumptions on  $u$ . Therefore, understanding the deviation in  $u(t, x)$  comes down to understanding conditional moments  $\mathbb{E}_{n-1}[\Delta x_\ell^b]$  of  $\Delta x_\ell$ . We may compute these moments using linear recurrence theory, in particular, the following lemma.

**Lemma 3.1.** *Let  $k$  be a natural number, and let  $a_n$  and  $b_n$  be sequences such that, for some*



constants  $c_j$ , the sequence  $a_n$  satisfies the  $k$ -step linear recurrence

$$a_n = b_n + \sum_{j=1}^k c_j a_{n-j}, \quad n \geq 1,$$

with initial data  $a_0, a_{-1}, \dots, a_{-k+1}$  prescribed. Then, for all  $n \geq 0$ ,

$$a_n = d_n + \sum_{j=1}^n G_{n-j} b_j,$$

where

1. the sequence  $G_n$  satisfies  $G_n = \sum_{i=1}^k c_i G_{n-i}$  for  $n \geq 1$ , with  $G_0 = 1$  and  $G_j = 0$  for  $-k+1 \leq j \leq -1$ ;
2. the sequence  $d_n$  satisfies  $d_n = \sum_{j=1}^k c_j d_{n-j}$  for  $n \geq 1$ , with  $d_0 = a_0$ ,  $d_{-1} = a_{-1}$ ,  $\dots$ ,  $d_{-k+1} = a_{-k+1}$ .

*Proof.* At  $n = 0$  the claim reads  $a_0 = d_0$ , which holds by construction, and the formula reproduces the prescribed initial data at negative indices, where the (empty) sum vanishes. Define the linear operator  $L$  by  $L(x)_n = x_n - \sum_{j=1}^k c_j x_{n-j}$ . Then  $L(a)_n = b_n$  and  $L(d)_n = 0$  for  $n \geq 1$ , and  $L(G)_n = \delta_{n,0}$  for  $n \geq 0$ , where  $\delta$  is the Kronecker delta. Extending the sum  $\sum_{j=1}^n G_{n-j} b_j$  by the convention  $G_m = 0$  for  $-k+1 \leq m \leq -1$ , we compute, for  $n \geq 1$ ,

$$L\left(\sum_{j \geq 1} G_{n-j} b_j\right)_n = \sum_{j=1}^n L(G)_{n-j} b_j = \sum_{j=1}^n \delta_{n-j,0} b_j = b_n.$$

Hence the right-hand side of the claimed formula satisfies the same recurrence as  $a_n$  for  $n \geq 1$ , with the same initial data, so the two sequences agree for all  $n \geq 0$ .  $\blacksquare$

In order to obtain a recurrence for the  $\Delta x_\ell$  to which Lemma 3.1 applies, we need the following criterion on a linear combination of the  $X_i$ .

**Lemma 3.2.** *For any random variables  $X_0, X_1, X_2, \dots, X_n$  and integer  $k < n$ , we can write*

$$X_n + \alpha_1 X_{n-1} + \dots + \alpha_k X_{n-k} = \Delta x_n + \sum_{i=1}^{k-1} \gamma_i \Delta x_{n-i}$$

for constants  $\gamma_i$  if and only if  $1 + \sum_{i=1}^k \alpha_i = 0$ , in which case  $\gamma_i = \sum_{j=0}^i \alpha_j$  (with  $\alpha_0 = 1$ ).

*Proof.* Expanding the right-hand side with  $\gamma_0 := 1$  gives

$$\sum_{i=0}^{k-1} \gamma_i (X_{n-i} - X_{n-i-1}) = \gamma_0 X_n + \sum_{i=1}^{k-1} (\gamma_i - \gamma_{i-1}) X_{n-i} - \gamma_{k-1} X_{n-k}.$$

Matching coefficients with the left-hand side requires  $\gamma_0 = 1 = \alpha_0$ ;  $\gamma_i - \gamma_{i-1} = \alpha_i$  for  $1 \leq i \leq k-1$ , forcing  $\gamma_i = \sum_{j \leq i} \alpha_j$ ; and  $-\gamma_{k-1} = \alpha_k$ , i.e.  $\sum_{j=0}^k \alpha_j = 0$ . Conversely, if  $\sum_j \alpha_j = 0$ , the coefficients  $\gamma_i = \sum_{j \leq i} \alpha_j$  satisfy all three sets of constraints. ■

The condition of Lemma 3.2 is exactly the consistency condition  $\rho(1) = 0$  of (A1). Applied to (1), it gives the recurrence

$$\Delta x_n = - \sum_{i=1}^{k-1} \gamma_i \Delta x_{n-i} + e_n, \quad \gamma_i = \sum_{j=0}^i \alpha_j, \quad (3)$$

valid for  $n \geq k$ , where

$$e_n = h \sum_{j=1}^k \beta_j f(X_{n-j}, t_{n-j}) + \sum_{j=1}^k \Gamma_j(X_{n-j}, t_{n-j}) I^{t_{n-j}, t_{n-j+1}}.$$

The characteristic polynomial of the reduced recurrence (3) is  $\rho(\zeta)/(\zeta - 1)$ . In particular, under the strict root condition of (A1), the kernel  $G_n$  obtained by applying Lemma 3.1 to (3) converges geometrically to the constant

$$G_\infty = \frac{1}{\rho'(1)},$$

and the differenced kernel  $G_n^* := G_n - G_{n-1}$  decays geometrically: there exist  $C_G > 0$  and  $\theta \in (0, 1)$  (the largest modulus among the roots of  $\rho(\zeta)/(\zeta - 1)$ ) with  $|G_n - G_\infty| + |G_n^*| \leq C_G \theta^n$ . Note that, because multiple stochastic integrals over  $[t_{n-1}, t_n]$  have zero mean given  $\mathcal{F}_{t_{n-1}}$ , and integrals over earlier subintervals are  $\mathcal{F}_{t_{n-1}}$ -measurable,

$$e_n \stackrel{\mathbb{E}}{=} h \sum_{j=1}^k \beta_j f(X_{n-j}, t_{n-j}) + \sum_{j=2}^k \Gamma_j(X_{n-j}, t_{n-j}) I^{t_{n-j}, t_{n-j+1}}.$$

By Lemma 3.1 applied to (3) we obtain the decomposition

$$\Delta x_n = d_n + \sum_{j=1}^n G_{n-j} e_j. \quad (4)$$

The moments  $\mathbb{E}_{n-1}[\Delta x_n^b]$  required in (2) are therefore mixed moments of the family  $\{e_j\}$ . Their computation requires care, because (a) the integrands  $q_j(X, \cdot)$  appearing in the  $e_j$  are random and adapted, and (b) the  $e_i$  and  $e_j$  with  $|i - j| < k$  share Wiener increments across different, overlapping subintervals. The systematic procedure, used repeatedly below and formalized in Appendix A, is: condition on the appropriate  $\sigma$ -algebra and use the tower property; Itô–Taylor expand each integrand around the left endpoint of its subinterval to the order required, so that—up to a remainder of controlled order—every term becomes an adapted coefficient multiplied by a multiple Wiener

integral with integrand 1 over a single subinterval; and compute the resulting mixed moments interval by interval, using independence of increments over disjoint intervals together with the product formula for iterated integrals. Moments of single multiple Itô integrals are supplied by the following identity.

**Theorem 3.3** (cf. [2, Corollary 5.2.4]). *Let  $\Delta W_h = W(t+h) - W(t)$ . Then the multiple Itô integral with  $\alpha$  ones in its subscript satisfies*

$$I_{1,1,\dots,1}^{t,t+h} = \frac{h^{\alpha/2}}{\alpha!} He_{\alpha}\left(\frac{\Delta W_h}{\sqrt{h}}\right),$$

where  $He_{\alpha}(x) = (-1)^{\alpha} e^{x^2/2} \frac{d^{\alpha}}{dx^{\alpha}} e^{-x^2/2}$  is the probabilists' Hermite polynomial.

For example,  $\alpha = 1$  gives  $\sqrt{h}(\Delta W_h/\sqrt{h}) = \Delta W_h$ , and  $\alpha = 2$  gives  $\frac{h}{2}((\Delta W_h/\sqrt{h})^2 - 1) = \frac{1}{2}(\Delta W_h^2 - h)$ , in agreement with  $I_{1,1}^{t,t+h} = \frac{1}{2}(\Delta W_h^2 - h)$ . In particular all moments of the  $I_{1,\dots,1}$  reduce to moments of the Gaussian  $\Delta W_h$ , and

$$(\mathbb{E} |I_{1,\dots,1}^{t,t+h}|^b)^{1/b} \leq C_{\alpha,b} h^{\alpha/2}. \quad (5)$$

Computing these moments leads us to our main theorem.

**Theorem 3.4.** *Let the SLMM (1) satisfy (A1), (A2), and (A4) with a starter of local weak error  $O(h^{\gamma+1})$ , and let (A3) hold with  $m = 2(r+2)$ , where  $\gamma$  is the weak order of consistency of the scheme. Then a weak error expansion*

$$\mathcal{E}(T, h) = c_0 h^{\gamma} + c_1 h^{\gamma+1} + \dots + c_r h^{\gamma+r} + O(h^{\gamma+r+1})$$

*may be computed explicitly and algorithmically; the coefficients  $c_i$  consist of integrals of expectations of smooth functionals of  $u$  and the coefficients along the true diffusion, plus starter-dependent boundary terms.*

*Proof.* See Appendix B. The proof is an induction on the expansion order: the conditional moments of  $\Delta x_n$  are expanded in  $h$  with adapted coefficients (Lemma A.3); the KB-PDE cancels the leading terms; the surviving terms of each order contain adapted factors involving past increments, which are re-expanded by the pairing estimates of Appendix A; and the strict root condition makes the resulting memory sums geometrically convergent, so that Riemann summation converts each local order into a global integral coefficient, with the initialization contributing geometrically localized boundary terms. ■

The strict root condition cannot be relaxed to the Dahlquist condition in Theorem 3.4: if  $\rho$  has a second simple root on the unit circle (e.g.  $\rho(\zeta) = \zeta^2 - 1$ ), the kernel  $G_n^*$  is bounded but does not

decay, the scheme carries a non-vanishing parasitic oscillation, and  $\mathcal{E}(T, h)$  admits no power-series expansion in  $h$ , exactly as in the deterministic theory of linear multistep methods.

The computation of moments  $\mathbb{E}_{n-1}[\Delta x_n^b]$  increases quickly in complexity as  $b$  increases, so one may choose to instead bound these moments from above. This can be done through Minkowski's inequality, applied directly to the decomposition (4).

**Lemma 3.5.** *Under (A1)–(A3), write each  $e_j$  as a finite sum of a drift term  $h p_j$  and terms of the form  $q \cdot I_{1,\dots,1}^{t_{j-i}, t_{j-i+1}}$  ( $\alpha$  ones), where  $p_j$  and  $q$  are adapted to  $\mathcal{F}_{t_{j-1}}$  and  $\mathcal{F}_{t_{j-i}}$ , respectively, with bounded  $L_b$  norms. Then*

$$\begin{aligned} (\mathbb{E} |\Delta x_n|^b)^{1/b} &\leq (\mathbb{E} |d_n|^b)^{1/b} + \sum_{j=1}^n |G_{n-j}| (\mathbb{E} |e_j|^b)^{1/b}, \\ (\mathbb{E} |e_j|^b)^{1/b} &\leq h \|p_j\|_{L_b} + \sum C_{\alpha,b} h^{\alpha/2} \|q\|_{L_b}, \end{aligned}$$

the last sum running over the constituent terms of  $e_j$ . In particular  $(\mathbb{E} |\Delta x_n|^b)^{1/b} = O(h^{1/2})$  uniformly in  $n$ .

*Proof.* The first inequality is Minkowski's inequality in  $L_b$  applied to (4). For the second, it suffices to bound each constituent  $q I$  of  $e_j$ . Since  $q$  is  $\mathcal{F}_{t_{j-i}}$ -measurable and the iterated integral  $I = I_{1,\dots,1}^{t_{j-i}, t_{j-i+1}}$  is a functional of the increments of  $W$  after time  $t_{j-i}$ , the random variables  $q$  and  $I$  are independent, whence  $\mathbb{E} |qI|^b = \mathbb{E} |q|^b \mathbb{E} |I|^b$  and  $\|qI\|_{L_b} = \|q\|_{L_b} \|I\|_{L_b} \leq C_{\alpha,b} h^{\alpha/2} \|q\|_{L_b}$  by (5). The uniformity in  $n$  follows since  $\sum_j |G_{n-j}| \cdot h^{1/2}$ -type sums are controlled by  $\sup_n |G_n| < \infty$  over the  $O(k)$  indices where increments overlap and by the geometric decay of  $G^*$  in the telescoped form; alternatively, apply the same argument to the memoryless form (3) unrolled once. ■

## 4 Weak Order Expansion of Certain Multistep Schemes

We now apply the theory developed in Section 3 to previously studied SLMMs and derive weak error behavior for a specific class of SLMMs tailored to small noise SDEs.

### 4.1 2-Step Schemes

We will use the framework of Theorem 3.4 to explicitly compute the weak error expansions of the schemes in [3]: MAB2 and two-step Maruyama schemes. These methods were presented in Examples 2.1 and 2.2. Throughout this subsection we take (A3) with  $m = 6$ , so that  $u \in C^{1,6}$  with polynomially bounded spatial derivatives; the sums below start at  $n = 2$ , with  $X_1$  produced by a starter as in (A4), contributing  $O(h^2)$  to the error.

In the case of MAB2, we will denote  $f_{n-1}^* = \frac{3}{2}f(X_{n-1}, t_{n-1}) - \frac{1}{2}f(X_{n-2}, t_{n-2})$  and  $g^* = g \partial_x g$ . Note that MAB2 has  $\alpha = (1, -1, 0)$ , so  $\gamma_1 = \alpha_0 + \alpha_1 = 0$  in (3) and the reduced recurrence is

memoryless:  $\Delta x_n = e_n$  exactly. Before we compute the weak order expansion, we will need to compute the conditional moments of  $\Delta x_n$ .

**Lemma 4.1.** *For the MAB2 scheme, with all coefficient functions evaluated at  $(X_{n-1}, t_{n-1})$  except  $f_{n-1}^*$  as above, the conditional moments given  $\mathcal{F}_{t_{n-1}}$  are*

$$\begin{aligned}\mathbb{E}_{n-1}[\Delta x_n] &= h f_{n-1}^*, \\ \mathbb{E}_{n-1}[\Delta x_n^2] &= h g^2 + h^2 \left( f_{n-1}^{*2} + \frac{1}{2} g^{*2} \right), \\ \mathbb{E}_{n-1}[\Delta x_n^3] &= 3h^2 g^2 (f_{n-1}^* + g^*) + R_{3,n}, \\ \mathbb{E}_{n-1}[\Delta x_n^4] &= 3h^2 g^4 + R_{4,n}, \\ \mathbb{E}_{n-1}[\Delta x_n^r] &= R_{r,n}, \quad r \geq 5,\end{aligned}$$

where each remainder satisfies  $\mathbb{E}|R_{\cdot,n}| \leq Ch^3$  with  $C$  depending only on  $k$ ,  $\|f\|_{C_b^1}$ ,  $\|g\|_{C_b^1}$ , and the moment bounds of (A3)–(A4).

*Proof.* Write  $\Delta x_n = A + B$  with  $A = h f_{n-1}^*$ , which is  $\mathcal{F}_{t_{n-1}}$ -measurable, and  $B = g I_1 + g^* I_{1,1}$ , where  $I_1 = I_1^{t_{n-1}, t_n} = \Delta W_n$  and, by Theorem 3.3,  $I_{1,1} = I_{1,1}^{t_{n-1}, t_n} = \frac{1}{2}(\Delta W_n^2 - h)$ . Conditionally on  $\mathcal{F}_{t_{n-1}}$ ,  $\Delta W_n$  is a centered Gaussian of variance  $h$ , independent of all coefficients, so every required moment reduces to a Gaussian moment. Using  $\mathbb{E}[\Delta W_n^{2m}] = (2m-1)!! h^m$  and  $\mathbb{E}[\Delta W_n^{2m+1}] = 0$ , direct computation gives

$$\begin{aligned}\mathbb{E}[I_{1,1}] &= 0, & \mathbb{E}[I_1 I_{1,1}] &= 0, & \mathbb{E}[I_{1,1}^2] &= \frac{h^2}{2}, \\ \mathbb{E}[I_1^2 I_{1,1}] &= h^2, & \mathbb{E}[I_1 I_{1,1}^2] &= 0, & \mathbb{E}[I_{1,1}^3] &= h^3.\end{aligned}$$

*First moment:*  $\mathbb{E}_{n-1}[\Delta x_n] = A$  since  $\mathbb{E}_{n-1}[B] = 0$ . *Second:*  $\mathbb{E}_{n-1}[\Delta x_n^2] = A^2 + 2A \mathbb{E}_{n-1}[B] + \mathbb{E}_{n-1}[B^2] = h^2 f_{n-1}^{*2} + g^2 h + \frac{1}{2} g^{*2} h^2$ , the cross term  $2g g^* \mathbb{E}[I_1 I_{1,1}]$  vanishing. *Third:*  $\mathbb{E}_{n-1}[\Delta x_n^3] = A^3 + 3A \mathbb{E}_{n-1}[B^2] + \mathbb{E}_{n-1}[B^3]$ . Here  $A^3 = O(h^3)$ ;  $3A \mathbb{E}_{n-1}[B^2] = 3h f_{n-1}^* (g^2 h + O(h^2)) = 3h^2 f_{n-1}^* g^2 + O(h^3)$ ; and

$$\mathbb{E}_{n-1}[B^3] = 3g^2 g^* \mathbb{E}[I_1^2 I_{1,1}] + 3g g^{*2} \mathbb{E}[I_1 I_{1,1}^2] + g^3 \mathbb{E}[I_1^3] + g^{*3} \mathbb{E}[I_{1,1}^3] = 3h^2 g^2 g^* + O(h^3).$$

*Fourth:*  $\mathbb{E}_{n-1}[B^4] = g^4 \mathbb{E}[I_1^4] + O(h^3) = 3g^4 h^2 + O(h^3)$ , since every term containing at least one factor  $g^* I_{1,1}$  or  $A$  contributes  $\mathbb{E}[I_1^3 I_{1,1}] = \frac{3}{2} h^3$  or higher powers of  $h$ ; the binomial terms  $4A \mathbb{E}_{n-1}[B^3]$ ,  $6A^2 \mathbb{E}_{n-1}[B^2]$ ,  $4A^3 \mathbb{E}_{n-1}[B]$ ,  $A^4$  are all  $O(h^3)$ . *Higher:* for  $r \geq 5$ , each monomial contains at least five factors from  $\{h, I_1, I_{1,1}\}$ ; every product of Gaussian factors of total polynomial degree  $\geq 5$  in  $\Delta W_n$  (or lower degree with compensating powers of  $h$ ) has expectation  $O(h^3)$ : odd total degrees vanish, and even total degrees  $\geq 6$  give  $O(h^3)$  by the moment scaling  $\mathbb{E}[\Delta W_n^{2m}] = O(h^m)$ . The remainders  $R_{\cdot,n}$  are products of  $h^3$  with polynomials in  $f_{n-1}^*, g, g^*$  evaluated along the scheme, so  $\mathbb{E}|R_{\cdot,n}| \leq Ch^3$  by the moment bounds of (A3)–(A4). We note that  $I_{1,1}$  is not Gaussian, so these

computations rest on Theorem 3.3 rather than on the moment generating function of a normal distribution.  $\blacksquare$

This lemma allows us to compute the weak error expansion. Its proof uses two auxiliary estimates proved in Appendix A: the pairing bound of Lemma A.2 (expectations of past increments against smooth adapted functionals are  $O(h)$ ) and the uniform first-order accuracy bound of Lemma A.4.

**Theorem 4.2.** *Under the assumptions above, the MAB2 scheme satisfies the weak error expansion*

$$\mathcal{E}(T, h) = c_0 h + O(h^2), \quad c_0 = \int_0^T \mathbb{E} \left[ \bar{\Phi}(s, X(s)) + \frac{1}{2} \Theta(s, X(s)) \right] ds,$$

where, writing  $g^* = g \partial_x g$  and evaluating all functions at  $(s, X(s))$  along the true diffusion,

$$\begin{aligned} \bar{\Phi} &= f \partial_{tx} u + \frac{1}{2} \partial_{tt} u + \left( \frac{1}{2} f^2 + \frac{1}{4} g^{*2} \right) \partial_{xx} u + \frac{1}{2} g^2 \partial_{txx} u + \frac{1}{2} g^2 (f + g^*) \partial_{xxx} u + \frac{1}{8} g^4 \partial_{xxxx} u, \\ \Theta &= (\partial_t f + f \partial_x f + \frac{1}{2} g^2 \partial_{xx} f) \partial_x u + g^2 \partial_x f \partial_{xx} u. \end{aligned}$$

*Proof.* Let  $u(t, x)$  solve the KB-PDE for  $\phi$ . Decomposing the error over the steps and isolating the starter,

$$\mathcal{E}(T, h) = \sum_{n=2}^N \mathbb{E} [u(t_n, X_{n-1} + \Delta x_n) - u(t_{n-1}, X_{n-1})] + \mathbb{E} [u(t_1, X_1) - u(t_0, X_0) + u(0, X_0) - u(0, X_0)],$$

where the starter term is  $O(h^2)$  by (A4) and the martingale property of  $u(t, X(t))$  under the true dynamics; more precisely,  $\mathbb{E}[u(t_1, X_1)] - \mathbb{E}[u(t_1, X(t_1))] = O(h^2)$  is the definition of local weak order 2, and  $\mathbb{E}[u(t_1, X(t_1))] = \mathbb{E}[u(0, X_0)]$ .

*Step 1: local expansion.* Taylor expanding at  $(t_{n-1}, X_{n-1})$  to total order 3 in  $(h, \Delta x_n)$  with fourth-order spatial term,

$$\begin{aligned} u(t + h, x + \Delta x_n) - u(t, x) &\stackrel{\mathbb{E}}{=} h \partial_t u + \Delta x_n \partial_x u + \frac{1}{2} (\Delta x_n)^2 \partial_{xx} u + h \Delta x_n \partial_{tx} u + \frac{1}{2} h^2 \partial_{tt} u \\ &\quad + \frac{1}{2} h (\Delta x_n)^2 \partial_{txx} u + \frac{1}{6} (\Delta x_n)^3 \partial_{xxx} u + \frac{1}{24} (\Delta x_n)^4 \partial_{xxxx} u + \varrho_n, \end{aligned}$$

where the Taylor remainder  $\varrho_n$  consists of terms  $h^a (\Delta x_n)^b$  with  $2a + b \geq 6$  or  $a \geq 2, b \geq 1$ , multiplied by bounded-in-expectation derivatives of  $u$  along the segment; by Lemma 3.5 and (A3),  $\mathbb{E}|\varrho_n| \leq Ch^3$ . Substituting the conditional moments of Lemma 4.1 and grouping by powers of  $h$ ,

$$\mathbb{E}[u(t_n, X_n) - u(t_{n-1}, X_{n-1})] = \mathbb{E} \left[ h \left( \partial_t u + f_{n-1}^* \partial_x u + \frac{1}{2} g^2 \partial_{xx} u \right) + h^2 \Phi_n \right] + O(h^3),$$

where  $\Phi_n$  equals  $\bar{\Phi}$  with  $f$  replaced by  $f_{n-1}^*$  in the  $\partial_{tx} u$ ,  $\partial_{xx} u$ , and  $\partial_{xxx} u$  coefficients, all evaluated at  $(t_{n-1}, X_{n-1})$ . The KB-PDE gives  $\partial_t u + f \partial_x u + \frac{1}{2} g^2 \partial_{xx} u = 0$  pointwise, so the order- $h$  bracket

reduces to the leftover

$$h(f_{n-1}^* - f_{n-1}) \partial_x u(t_{n-1}, X_{n-1}) = \frac{h}{2}(f_{n-1} - f_{n-2}) \partial_x u(t_{n-1}, X_{n-1}),$$

with  $f_j := f(X_j, t_j)$ .

*Step 2: the leftover term.* This term does *not* telescope, because the weight  $\partial_x u(t_{n-1}, X_{n-1})$  varies with  $n$ ; moreover its expectation is of order  $h$  rather than  $o(h)$ , because  $f_{n-1} - f_{n-2}$  contains the increment  $\partial_x f \cdot g \Delta W_{n-1}$ , which is correlated with  $\partial_x u(t_{n-1}, X_{n-1})$ . We expand it. Condition at  $\mathcal{F}_{t_{n-2}}$  and expand both factors over one MAB2 step, with

$$\Delta X := X_{n-1} - X_{n-2} = hf_{n-2}^* + g \Delta W_{n-1} + g^* I_{1,1}^{t_{n-2}, t_{n-1}},$$

all coefficients at  $(t_{n-2}, X_{n-2})$ :

$$\begin{aligned} f_{n-1} - f_{n-2} &= \partial_x f \Delta X + \frac{1}{2} \partial_{xx} f (\Delta X)^2 + h \partial_t f + r_n^{(1)}, \\ \partial_x u(t_{n-1}, X_{n-1}) &= \partial_x u + \partial_{xx} u \Delta X + h \partial_{tx} u + r_n^{(2)}, \end{aligned}$$

with  $\mathbb{E}|r_n^{(i)}|^2 \leq Ch^3$  by Lemma 3.5. Taking  $\mathbb{E}_{n-2}$  of the product and using the moments of Lemma 4.1 (for the step  $n-1$ ): the mean parts contribute

$$h(\partial_t f + f_{n-2}^* \partial_x f + \frac{1}{2} g^2 \partial_{xx} f) \partial_x u + O(h^2);$$

the covariance of the two  $\Delta X$  factors contributes

$$\mathbb{E}_{n-2}[\partial_x f \Delta X \cdot \partial_{xx} u \Delta X] = \partial_x f \partial_{xx} u \mathbb{E}_{n-2}[(\Delta X)^2] = h g^2 \partial_x f \partial_{xx} u + O(h^2);$$

and all remaining pairings are  $O(h^2)$ : odd Gaussian moments vanish,  $\mathbb{E}[\Delta W I_{1,1}] = 0$ , products involving  $r_n^{(i)}$  are  $O(h^{3/2}) \cdot O(h^{1/2}) = O(h^2)$  by Cauchy–Schwarz, and the term  $\frac{1}{2} \partial_{xx} f (\Delta X)^2 \cdot \partial_{xx} u \Delta X$  is a third-moment term of size  $O(h^2)$ . Finally,  $f_{n-2}^* - f_{n-2} = \frac{1}{2}(f_{n-2} - f_{n-3})$  is  $\mathcal{F}_{t_{n-2}}$ -measurable, of size  $O(h^{1/2})$  in  $L_2$ , and multiplies the smooth adapted factor  $h \partial_x f \partial_x u$ ; by the pairing bound of Lemma A.2, its expectation contributes  $h \cdot O(h) = O(h^2)$ . Collecting,

$$\frac{h}{2} \mathbb{E}[(f_{n-1} - f_{n-2}) \partial_x u(t_{n-1}, X_{n-1})] = \frac{h^2}{2} \mathbb{E}[\Theta(t_{n-2}, X_{n-2})] + O(h^3).$$

*Step 3: summation.* Combining Steps 1–2,

$$\mathcal{E}(T, h) = h^2 \sum_{n=2}^N \mathbb{E} \left[ \Phi_n(t_{n-1}, X_{n-1}) + \frac{1}{2} \Theta(t_{n-2}, X_{n-2}) \right] + O(h^2).$$

Three replacements convert the sum into the stated integral, each with total cost  $O(h^2)$ . (i) Replace

$f_{n-1}^*$  by  $f_{n-1}$  inside  $\Phi_n$ : the difference is  $\frac{1}{2}(f_{n-1} - f_{n-2})$  times smooth adapted factors, whose expectation is  $O(h)$  per step by Lemma A.2, hence  $h^2 \cdot N \cdot O(h) = O(h^2)$  in total. (ii) Replace the scheme's marginals  $X_{n-1}$  by the diffusion's marginals  $X(t_{n-1})$ : the integrands  $\bar{\Phi} + \frac{1}{2}\Theta$  are, for each fixed time, functions in  $C^4$  with polynomially bounded derivatives, uniformly in the time index by (A3); by the uniform first-order accuracy bound of Lemma A.4 (whose proof uses only Steps 1–2 above with a generic terminal function, and is therefore not circular), each replacement costs  $O(h)$ , for a total of  $h^2 \cdot N \cdot O(h) = O(h^2)$ . (iii) Replace the Riemann sum by the integral: the map  $s \mapsto \mathbb{E}[(\bar{\Phi} + \frac{1}{2}\Theta)(s, X(s))]$  is Lipschitz on  $[0, T]$  by (A3) and Itô's formula, so the Riemann-sum error is  $O(h)$  on the prefactor  $h$ , i.e.  $O(h^2)$ . This yields  $\mathcal{E}(T, h) = h \int_0^T \mathbb{E}[\bar{\Phi} + \frac{1}{2}\Theta](s, X(s)) ds + O(h^2)$ . ■

Due to the boundedness of partial derivatives of  $u$ , we have the following corollary.

**Corollary 4.3.** *Under the assumptions of Theorem 4.2, the MAB2 scheme has weak order 1.*

We have a similar result for two-step Maruyama schemes under appropriate consistency conditions. Notice that  $X_n = -\alpha_1 X_{n-1} - \alpha_2 X_{n-2} + e_n$ , where now

$$e_n = h(\beta_1 f_{n-1} + \beta_2 f_{n-2}) + \gamma_1 g_{n-1} \Delta W_n + \gamma_2 g_{n-2} \Delta W_{n-1},$$

with  $f_j = f(X_j, t_j)$ ,  $g_j = g(X_j, t_j)$ , and  $\Delta W_j = W(t_j) - W(t_{j-1})$ : the diffusion term with coefficient  $\gamma_2$  carries the increment  $I_1^{t_{n-2}, t_{n-1}} = \Delta W_{n-1}$ , not  $\Delta W_n$ . (We define  $f$  and  $g$  at negative indices to be 0.) By  $\rho(1) = 0$ , the second root of  $\rho(\zeta) = \zeta^2 + \alpha_1 \zeta + \alpha_2$  is  $\zeta_2 = \alpha_2$ , and the reduced recurrence (3) reads

$$\Delta x_n = \zeta_2 \Delta x_{n-1} + e_n. \quad (6)$$

We may also express  $X_n$  explicitly by Lemma 3.1:  $X_n = d_n + \sum_{j=1}^n G_{n-j} e_j$ . Let  $G_n^* = G_n - G_{n-1}$  for  $n \geq 1$  and  $G_0^* = 1$ , and let  $\Lambda_n^f = \sum_{j=1}^n G_{n-j}^* (\beta_1 f_{j-1} + \beta_2 f_{j-2})$ . Grouping the noise contributions by increment rather than by step yields the following representation.

**Lemma 4.4.** *For explicit two-step Maruyama schemes,*

$$\Delta x_n = (d_n - d_{n-1}) + h \Lambda_n^f + \sum_{j=1}^n c_j^{(n)} g_{j-1} \Delta W_j, \quad c_j^{(n)} := \gamma_1 G_{n-j}^* + \gamma_2 G_{n-j-1}^*,$$

with the convention  $G_m^* = 0$  for  $m < 0$ . Under (A1),  $|c_j^{(n)}| \leq C\theta^{n-j}$  for some  $\theta \in (0, 1)$ , and here  $\theta = |\zeta_2|$ ,  $G_m^* = \zeta_2^m$ ,  $G_\infty = 1/(1 - \zeta_2) = 1/\rho'(1)$ .

*Proof.* Substitute the expression for  $e_j$  into  $X_n = d_n + \sum_{j \leq n} G_{n-j} e_j$  and collect, for each fixed increment  $\Delta W_j$ , the two step indices at which it appears, namely step  $j$  (weight  $\gamma_1 G_{n-j}$ ) and step  $j+1$  (weight  $\gamma_2 G_{n-j-1}$ ):

$$X_n = d_n + h \sum_{j \leq n} G_{n-j} (\beta_1 f_{j-1} + \beta_2 f_{j-2}) + \sum_{j \leq n} (\gamma_1 G_{n-j} + \gamma_2 G_{n-j-1}) g_{j-1} \Delta W_j.$$



Differencing consecutive  $n$  gives the claim. For the kernel: from (6),  $G_n = \sum_{i=0}^n \zeta_2^i$ , so  $G_n^* = \zeta_2^n$ , and  $G_\infty = 1/(1 - \zeta_2)$ ; since  $\rho(\zeta) = (\zeta - 1)(\zeta - \zeta_2)$ ,  $\rho'(1) = 1 - \zeta_2$ . ■

The representation yields moments via martingale orthogonality: since  $g_{j-1}$  is  $\mathcal{F}_{t_{j-1}}$ -measurable and  $\Delta W_j$  is independent of  $\mathcal{F}_{t_{j-1}}$ , cross terms between distinct increments vanish.

**Lemma 4.5.** *For explicit two-step Maruyama schemes,*

$$\begin{aligned}\mathbb{E}[\Delta x_n] &= \mathbb{E}[d_n - d_{n-1}] + h \mathbb{E}[\Lambda_n^f], \\ \mathbb{E}[\Delta x_n^2] &= \mathbb{E}\left[(d_n - d_{n-1} + h \Lambda_n^f)^2\right] + h \sum_{j=1}^n (c_j^{(n)})^2 \mathbb{E}[g_{j-1}^2].\end{aligned}$$

*Proof.* The first identity holds since each  $\Delta W_j$  has mean zero and is independent of  $g_{j-1}$ . For the second, expand the square; the cross terms between the drift part and the noise sum vanish term by term, since for the term indexed  $j$  the factor  $\Delta W_j$  has zero mean conditionally on the  $\sigma$ -algebra generated by  $\mathcal{F}_{t_{j-1}}$  and the increments  $(\Delta W_i)_{i \neq j}$ , with respect to which all other factors in the product are measurable except through later increments, which are themselves independent and mean-zero; a cleaner route is to condition successively on  $\mathcal{F}_{t_{j-1}}$  for the largest index  $j$  appearing. Cross terms between distinct noise indices  $j < j'$  vanish by conditioning on  $\mathcal{F}_{t_{j'-1}}$ . The diagonal terms give  $\mathbb{E}[(c_j^{(n)} g_{j-1} \Delta W_j)^2] = h (c_j^{(n)})^2 \mathbb{E}[g_{j-1}^2]$  by independence. ■

We emphasize that these are *unconditional* moments, and that conditionally on  $\mathcal{F}_{t_{n-1}}$  the past increments  $\Delta W_j$ ,  $j \leq n-1$ , appearing in Lemma 4.4 are known quantities; the pairing of powers of  $\Delta x_n$  with derivatives of  $u$  at  $(t_{n-1}, X_{n-1})$  in (2) therefore requires the conditional structure, which is what the proof of the following theorem uses. (This is also why the MAB2 analysis is comparatively clean: there  $\gamma_1^{(3)} = \alpha_0 + \alpha_1 = 0$ , so  $\Delta x_n = e_n$  has no memory.)

**Theorem 4.6.** *Let an explicit two-step Maruyama scheme satisfy (A1) (equivalently here,  $\rho(1) = 0$  and  $|\alpha_2| < 1$ ), with (A3) at  $m = 6$  and a starter as in (A4). Then:*

- (i) *if  $\sum_j \beta_j = \rho'(1)$  and  $(\sum_j \gamma_j)^2 = \rho'(1)^2$ , the scheme has weak order 1;*
- (ii) *conversely, if the scheme has weak order 1 (for all  $f, g, \phi$  as in (A3)), then  $\sum_j \beta_j = \rho'(1)$  and  $(\sum_j \gamma_j)^2 = \rho'(1)^2$ .*

Here  $\rho'(1) = 1 - \alpha_2 = -\sum_j j \alpha_j$ .

*Proof.* See Appendix C. ■

*Remarks on Theorem 4.6.* The drift relation coincides with the mean-square consistency condition of Buckwar–Winkler [3]. The noise relation is the *weak* form of theirs: mean-square consistency forces  $\sum_j \gamma_j = \rho'(1)$  with the positive sign, because strong convergence compares the scheme

pathwise with the same driving Brownian motion; weak convergence sees the noise weights only through even functionals at this order, and indeed  $\gamma \mapsto -\gamma$  leaves the law of the scheme invariant (as  $-W$  is again a Brownian motion), so only  $(\sum \gamma)^2$  can be pinned down. We also note that the strict root condition cannot be weakened to the Dahlquist condition:  $\rho(\zeta) = \zeta^2 - 1$  satisfies Dahlquist (simple roots  $\pm 1$ ) but  $G_n^* = (-1)^n$  does not decay, the scheme carries a non-vanishing parasitic mode, and the expansion underlying the proof fails.

## 4.2 Multistep Adams-Bashforth

We now analyze a particular class of SLMMs tailored to the small-noise regime. Buckwar and Winkler show in [3] that if the noise term in the original SDE is small, SLMMs provide asymptotic improvement over classical one-step methods in mean square convergence error. More precisely, we consider SDEs of the form

$$dX_t = f(X_t, t)dt + g(X_t, t)dW_t,$$

where  $g = \varepsilon \hat{g}$  for some small constant  $\varepsilon$  and some function  $\hat{g} \in C_b^r$  for a given  $r$ . Buckwar and Winkler only consider 2-step Maruyama schemes, which we address with Theorem 4.6. We instead study SLMMs that adapt the Adams-Bashforth method to better capture the drift term. We obtain small noise asymptotics for such schemes using a continuous version of the algorithm in Theorem 3.4.

We may define the  $k$ -step Adams-Bashforth SLMM,  $k$ -ABM for short, as follows.

**Definition 4.7.** Let  $P_k(t)$  and  $Q_k(t)$  be the Lagrange interpolating polynomials through the points  $(t_{n-j}, f(X_{n-j}, t_{n-j}))$  and  $(t_{n-j}, g(X_{n-j}, t_{n-j}))$  for  $1 \leq j \leq k$ , respectively. The  $k$ -ABM scheme is the SLMM defined by

$$X_{n+1} = X_n + \int_{[t_n, t_{n+1}]} P_k(t) dt + \int_{[t_n, t_{n+1}]} Q_k(t) dW_t.$$

$P_k$  and  $Q_k$  are  $\mathcal{F}_{t_n}$ -measurable random polynomials, and the scheme *extrapolates* them to  $[t_n, t_{n+1}]$ , outside the interpolation nodes, exactly as Adams-Bashforth does. The relevant elementary bounds are collected in the following lemma, proved in Appendix D.

**Lemma 4.8.** *Let*

$$\ell_j(t) = \prod_{i \neq j} \frac{t - t_{n-i}}{t_{n-j} - t_{n-i}}, \quad 1 \leq j \leq k,$$

*be the Lagrange basis on the nodes  $t_{n-1}, \dots, t_{n-k}$ . Then, for  $t \in [t_n, t_{n+1}]$ : (a)  $\prod_{j=1}^k |t - t_{n-j}| \leq (k+1)! h^k$ ; (b)  $C_k := \sup_{t \in [t_n, t_{n+1}]} \sum_{j=1}^k |\ell_j(t)| \leq k(k+1)!$ ; and (c) for any function  $\psi \in C^{k+1}([t_{n-k}, t_{n+1}])$ , the interpolant  $L_k \psi$  on these nodes satisfies  $\sup_{t \in [t_n, t_{n+1}]} |\psi(t) - L_k \psi(t)| \leq \|\psi^{(k+1)}\|_\infty h^k$ .*

The proof of the main theorem below rests on a small-noise decomposition of both the SDE and the scheme, stated next and proved in Appendix D. An earlier version of this paper attempted

instead to apply the Lagrange bound to the map  $t \mapsto f(\mathbb{E}_n \tilde{X}_t, t)$ ; but for  $t \leq t_n$  one has  $\mathbb{E}_n \tilde{X}_t = \tilde{X}_t$ , the *realized* path, which contains realized Wiener integrals and is almost surely nowhere differentiable, so that bound does not apply. The decomposition replaces it.

**Lemma 4.9.** *Assume  $f \in C_b^{k+1}(\mathbb{R} \times I)$  jointly,  $\hat{g} \in C_b^{k+1}$ , and (A4). Let  $x^0$  solve  $\dot{x}_t^0 = f(x_t^0, t)$ ,  $x_0^0 = X_0$ , and let  $R$  solve the linear SDE  $dR_t = \partial_x f(x_t^0, t) R_t dt + \hat{g}(x_t^0, t) dW_t$ ,  $R_0 = 0$ , so that  $\mathbb{E} R_t \equiv 0$ . Then:*

- (a)  $\sup_{t \leq T} \|X(t) - x_t^0 - \varepsilon R_t\|_{L_2} \leq C\varepsilon^2$ ;
- (b) *there is a deterministic sequence  $x_n^{0,h}$  (the  $k$ -step Adams–Bashforth iterates for  $\dot{x} = f$ ) with  $\max_n |x_n^{0,h} - x_{t_n}^0| \leq Ch^k$ , and a mean-zero process  $R_n^h$  adapted to  $\mathcal{F}_{t_n}$ , such that the  $k$ -ABM iterates satisfy  $\max_n \|X_n - x_n^{0,h} - \varepsilon R_n^h\|_{L_2} \leq C(\varepsilon^2 + \varepsilon h^k)$  and  $\max_n \|R_n^h\|_{L_4} \leq C$ ;*
- (c) *consequently  $f(X_{n-j}, t_{n-j}) = F(t_{n-j}) + \varepsilon \partial_x f(x_{t_{n-j}}^0, t_{n-j}) R_{n-j}^h + \eta_{n-j}$  with  $F(t) = f(x_t^0, t) \in C^{k+1}(I)$  and  $\|\eta_{n-j}\|_{L_2} \leq C(\varepsilon^2 + h^k)$ , and analogously for  $\hat{g}^2$  with  $G(t) = \hat{g}^2(x_t^0, t)$ .*

Our main result is the following estimate of the weak error for the  $k$ -ABM.

**Theorem 4.10.** *Assume  $f \in C_b^{k+1}(\mathbb{R} \times I)$  jointly in  $(x, t)$ ,  $\hat{g} \in C_b^{k+1}$ , and  $\phi \in C^4$  with polynomially bounded derivatives. Then, with  $F, G$  as in Lemma 4.9,*

$$\begin{aligned} \text{Err}(T, h) &\leq CT \left[ h\varepsilon^2 C_k^2 \|\hat{g}\|_\infty^2 \left( \|\partial_x u\|_\infty \|\partial_{xx} f\|_\infty + \varepsilon^2 \|\partial_{xx} u\|_\infty \|\partial_{xx}(\hat{g}^2)\|_\infty \right) \right. \\ &\quad \left. + h^k \left( \|\partial_x u\|_\infty \|F^{(k+1)}\|_\infty + \varepsilon^2 \|\partial_{xx} u\|_\infty \|G^{(k+1)}\|_\infty \right) \right] \\ &\quad + CT(h\varepsilon^3 + h^k(\varepsilon + h) + h^{2k-1}). \end{aligned}$$

In particular,  $\text{Err}(T, h) = O(h\varepsilon^2 + h^k)$ .

*Proof.* Let  $u(t, x)$  solve the KB-PDE and set  $e_i = \mathbb{E}[u(t_{i+1}, X_{i+1}) - u(t_i, X_i)]$ , so that  $\mathcal{E}(T, h) = \sum_{i \geq k} e_i + O(h^2)$ , the starter steps contributing  $O(h^2)$  as in Theorem 4.2. Interpolate continuously:

$$\tilde{X}_t = X_n + \int_{t_n}^t P_k(s) ds + \int_{t_n}^t Q_k(s) dW_s, \quad t \in [t_n, t_{n+1}].$$

By Itô's Lemma applied to  $u(t, \tilde{X}_t)$  (the stochastic integral term has zero expectation by the boundedness of  $\partial_x u$ ,  $Q_k$ , and the moment bounds), and by the KB-PDE  $(\partial_t + \mathcal{L})u = 0$  evaluated at  $(t, \tilde{X}_t)$ ,

$$e_i = \mathbb{E} \int_{[t_i, t_{i+1}]} \left[ \partial_x u(t, \tilde{X}_t) (P_k(t) - f(\tilde{X}_t, t)) + \frac{1}{2} \partial_{xx} u(t, \tilde{X}_t) (Q_k(t)^2 - g(\tilde{X}_t, t)^2) \right] dt. \quad (7)$$

We treat the drift term; the diffusion term is identical with  $(f, \partial_x u)$  replaced by  $(g^2, \frac{1}{2} \partial_{xx} u)$  and carries an overall factor  $\varepsilon^2$  since  $g^2 = \varepsilon^2 \hat{g}^2$ . Decompose, for  $t \in [t_i, t_{i+1}]$ ,

$$f(\tilde{X}_t, t) - P_k(t) = \underbrace{(f(\tilde{X}_t, t) - f(\mathbb{E}_i \tilde{X}_t, t))}_{(I)} + \underbrace{(f(\mathbb{E}_i \tilde{X}_t, t) - P_k(t))}_{(II)},$$

where  $\mathbb{E}_i \tilde{X}_t = X_i + \int_{t_i}^t P_k ds$  for  $t \in [t_i, t_{i+1}]$  (the conditional mean on the *current* interval, which is well defined and smooth in  $t$ ).

*Term (I): fluctuation on the current interval.* Given  $\mathcal{F}_{t_i}$ ,  $M_t := \tilde{X}_t - \mathbb{E}_i \tilde{X}_t = \int_{t_i}^t Q_k dW$  is a centered Gaussian with variance  $\int_{t_i}^t Q_k^2 ds \leq (t - t_i) \sup |Q_k|^2$ . By Lemma 4.8(b),  $\sup_{[t_i, t_{i+1}]} |Q_k| \leq C_k \varepsilon \|\hat{g}\|_\infty$ . Taylor expanding  $f$  about the conditional mean  $m_t = \mathbb{E}_i \tilde{X}_t$  and taking  $\mathbb{E}_i$ : the linear term vanishes; the quadratic term gives  $\frac{1}{2} \partial_{xx} f(m_t, t) \int_{t_i}^t Q_k^2 ds$ ; and the remainder is bounded by  $\frac{1}{24} \|\partial_x^4 f\|_\infty \mathbb{E}_i[M_t^4] \leq C(t - t_i)^2 C_k^4 \varepsilon^4 \|\hat{g}\|_\infty^4$ , using  $\mathbb{E}_i[M_t^3]$ 's contribution being controlled through the fourth moment  $\mathbb{E}_i[M_t^4] = 3(\int Q_k^2)^2$  after pairing with the bounded third derivative (the conditionally Gaussian third moment vanishes exactly). Pairing with  $\partial_x u(t, \tilde{X}_t)$  costs only the sup norm. Integrating over  $[t_i, t_{i+1}]$ ,

$$\left| \mathbb{E} \int_{[t_i, t_{i+1}]} \partial_x u(I) dt \right| \leq \frac{1}{2} h^2 \|\partial_x u\|_\infty \|\partial_{xx} f\|_\infty C_k^2 \varepsilon^2 \|\hat{g}\|_\infty^2 + C h^3 \varepsilon^4.$$

*Term (II): interpolation of the drift.* By Lemma 4.9(c), the interpolated data split as

$$f(X_{i-j}, t_{i-j}) = F(t_{i-j}) + \varepsilon \partial_x f(x^0) R_{i-j}^h + \eta_{i-j},$$

so by linearity of Lagrange interpolation  $P_k = L_k F + \varepsilon L_k [\partial_x f(x^0) R^h] + L_k \eta$ , and correspondingly, writing  $m_t = x_t^0 + \varepsilon \bar{R}_t + O_{L_2}(\varepsilon^2 + h^k)$  with  $\bar{R}_t$  the conditional mean of the fluctuation (an  $\mathcal{F}_{t_i}$ -measurable, mean-zero random variable by Lemma 4.9(b) and  $\mathbb{E} R^h = 0$ ),

$$(II) = \underbrace{(F(t) - L_k F(t))}_{(IIa)} + \varepsilon \underbrace{(\partial_x f(x_t^0, t) \bar{R}_t - L_k [\partial_x f(x^0) R^h](t))}_{(IIb)} + (IIc),$$

where  $\|(IIc)\|_{L_1} \leq C C_k (\varepsilon^2 + h^k (\varepsilon + h))$  collects the  $\eta$ -terms, their interpolants (Lemma 4.8(b)), and the quadratic Taylor remainders in the fluctuation of size  $O(\varepsilon^2)$ . For (IIa):  $F \in C^{k+1}$ , so Lemma 4.8(c) gives  $|(IIa)| \leq \|F^{(k+1)}\|_\infty h^k$ ; pairing with  $\|\partial_x u\|_\infty$  and integrating gives the  $h^{k+1}$  per-step contribution. For (IIb): it is a linear combination, with  $\mathcal{F}_{t_i}$ -measurable bounded weights of size  $\leq C(1 + C_k)$ , of the mean-zero random variables  $R_{i-j}^h$  ( $1 \leq j \leq k$ ) and  $\bar{R}_t$ . Pair it with  $\partial_x u(t, \tilde{X}_t) = \partial_x u(t, x_t^0) + O_{L_2}(\varepsilon + h^k)$ : the deterministic factor  $\partial_x u(t, x_t^0)$  pulls out of the expectation and multiplies mean-zero quantities, contributing 0; the correction pairs an  $O_{L_2}(\varepsilon + h^k)$  factor with

an  $O_{L_2}(1)$  mean-zero factor, contributing  $O(\varepsilon + h^k)$ . Hence

$$\left| \mathbb{E} \int_{[t_i, t_{i+1}]} \partial_x u(\Pi) dt \right| \leq h \left( \|\partial_x u\|_\infty \|F^{(k+1)}\|_\infty h^k + C\varepsilon(\varepsilon + h^k) + C C_k(\varepsilon^2 + h^k(\varepsilon + h)) \right).$$

The diffusion term in (7) is handled identically with  $G$  in place of  $F$  and the overall factor  $\varepsilon^2$ ; its fluctuation (I)-analogue is of size  $h^2\varepsilon^4$  per step.

Summing the  $T/h$  steps and collecting powers gives the displayed bound; the terms  $h\varepsilon^3$ ,  $h^k(\varepsilon + h)$ , and  $h^{2k-1}$  (the latter from  $(h^k)^2/h$ -type cross terms) are absorbed into the lower-order-terms line, and all are  $O(h\varepsilon^2 + h^k)$  by Young's inequality ( $\varepsilon^3 \leq \varepsilon^2$  for  $\varepsilon \leq 1$ ;  $h^k\varepsilon \leq \frac{1}{2}(h\varepsilon^2 + h^{2k-1})$ ). ■

Note that Theorem 4.10 shows that SLMs can asymptotically decrease the weak error of their analogous one-step schemes in the small noise case: for fixed small  $\varepsilon$ , the error is dominated by the floor  $h\varepsilon^2$  once  $h^k \lesssim h\varepsilon^2$ , i.e. below the crossover  $h^* \sim \varepsilon^{2/(k-1)}$ . We also note that the  $k$ -ABM shares its characteristic polynomial  $\rho(\zeta) = \zeta^{k-1}(\zeta - 1)$  with the classical Adams–Bashforth method, and hence is zero-stable for every  $k$ ; what deteriorates as  $k$  grows is the region of *absolute* stability, as in the ODE case [16]. One should also note that the infinity norms in Theorem 4.10 can be restricted to a neighborhood of the flow over  $I$ , and terms like  $\|\partial_x u\|_\infty$  and  $\|\partial_{xx} u\|_\infty$  can be upper bounded by quantities independent of  $u$  by the work in [17].

## 5 Lévy-driven SDEs

We now extend the weak error framework to Lévy-driven SDEs of the form given in Section 2,

$$dX_t = f(X_{t-}) dt + g(X_{t-}) dW_t + \int_{|z|<1} \gamma(X_{t-}, z) \tilde{J}(dt, dz) + \int_{|z|\geq 1} \delta(X_{t-}, z) J(dt, dz),$$

whose generator is the integro-differential operator

$$\begin{aligned} \mathcal{L}u(t, x) &= f(x) \partial_x u + \frac{1}{2} g(x)^2 \partial_{xx} u + \int_{|z|<1} [u(t, x + \gamma(x, z)) - u(t, x) - \gamma(x, z) \partial_x u(t, x)] \nu(dz) \\ &\quad + \int_{|z|\geq 1} [u(t, x + \delta(x, z)) - u(t, x)] \nu(dz). \end{aligned}$$

We work under the following hypotheses, in force throughout this section.

(L1) (*Moments of the jump part.*) There exist measurable dominating functions  $\rho_1$  on  $\{|z| < 1\}$  and  $\rho_2$  on  $\{|z| \geq 1\}$  with

$$\sup_x |\gamma(x, z)| \leq \rho_1(z), \quad \sup_x |\delta(x, z)| \leq \rho_2(z), \quad \int_{|z|<1} \rho_1^2 d\nu + \int_{|z|\geq 1} \rho_2^2 d\nu < \infty,$$

and additionally  $M_3 := \int_{|z|<1} \rho_1^3 d\nu + \int_{|z|\geq 1} \rho_2^3 d\nu < \infty$  where stated.

(L2) (*Regularity.*)  $f, g \in C_b^m$ ;  $\gamma(\cdot, z), \delta(\cdot, z) \in C_b^m$  in  $x$  for each  $z$ , with  $x$ -derivatives up to order  $m$  dominated by  $\rho_1(z), \rho_2(z)$  respectively; and  $\phi \in C^m$  with polynomially bounded derivatives. Under (L1)–(L2) the KB equation  $\partial_t u + \mathcal{L}u = 0$ ,  $u(T, \cdot) = \phi$ , has a solution  $u \in C^{1,m}$  with polynomially bounded spatial derivatives; see Protter–Talay [17] and [18, Chapter 6].

The finiteness of  $\int_{|z| \geq 1} \rho_2 d\nu$  follows automatically from (L1) and  $\nu(\{|z| \geq 1\}) < \infty$  by Cauchy–Schwarz.

The recurrence reduction of Section 3 is purely algebraic and carries over verbatim: Lemma 3.1, Lemma 3.2, and the decomposition (4) hold with  $e_n$  now including the jump approximation terms  $\Xi_j$ . What changes is the moment calculus: Gaussian pairings are replaced by pairings against compensated Poisson integrals, for which the natural integration-by-parts tool is the Mecke formula (Lemma E.1 in Appendix E). The analogue of Theorem 3.4 is the following.

**Proposition 5.1.** *Let a Lévy SLMM satisfy (A1), (A2), (A4), and let (L1)–(L2) hold with  $m = m(r)$  large enough, with the jump terms  $\Xi_j$  built from the increments  $J((t_{\ell-j}, t_{\ell-j+1}], dz)$  and  $\tilde{J}((t_{\ell-j}, t_{\ell-j+1}], dz)$  integrated against  $C_b^m$  coefficients dominated as in (L2). Then the scheme admits a computable weak error expansion*

$$\mathcal{E}(T, h) = c_0 h^\gamma + \cdots + c_r h^{\gamma+r} + O(h^{\gamma+r+1}),$$

with coefficients given by time integrals of expectations of functionals of  $u$  and the coefficients along the true process.

*Proof.* See Appendix E. The proof follows the induction of Appendix B, with two substitutions: the pairing estimates use the Mecke formula in place of Gaussian integration by parts, and the regularity of  $u$  is supplied by [17] rather than the diffusion theory. The inputs quoted from [17] and [18] are standard and are cited precisely where used. ■

An earlier version stated this result as a theorem with a one-paragraph sketch; since the argument, while parallel to Appendix B, relies on external regularity inputs, we prefer the label proposition and an explicit list of citations.

We now analyze the Lévy analogue of the  $k$ -ABM in the small-noise regime. As before  $g = \varepsilon \hat{g}$ , and we additionally scale the small-jump coefficient  $\gamma = \varepsilon \hat{\gamma}$  with  $\hat{\gamma}$  dominated by  $\rho_1$ ; the large jumps  $\delta$  are not scaled.

**Definition 5.2.** Let  $P_k, Q_k$  be as in Definition 4.7 (interpolating  $f$  and  $g$  at the nodes  $t_{n-j}$ ,  $1 \leq j \leq k$ ). The Lévy  $k$ -ABM is

$$\begin{aligned} X_{n+1} = X_n &+ \int_{[t_n, t_{n+1}]} P_k(t) dt + \int_{[t_n, t_{n+1}]} Q_k(t) dW_t \\ &+ \int_{[t_n, t_{n+1}]} \int_{|z| < 1} \gamma(X_n, z) \tilde{J}(dt, dz) + \int_{[t_n, t_{n+1}]} \int_{|z| \geq 1} \delta(X_n, z) J(dt, dz), \end{aligned}$$

i.e. Adams–Bashforth extrapolation for drift and diffusion, Euler treatment of the jumps.

For the local analysis we interpolate continuously on  $[t_i, t_{i+1}]$ , freezing the jump coefficients at  $X_i$ :

$$\tilde{X}_t = X_i + \int_{t_i}^t P_k ds + \int_{t_i}^t Q_k dW + \int_{t_i}^t \int_{|z|<1} \gamma(X_i, z) \tilde{J}(ds, dz) + \int_{t_i}^t \int_{|z|\geq 1} \delta(X_i, z) J(ds, dz).$$

Define the compensated jump fluctuation and its quadratic intensity

$$J_t^* = \int_{t_i}^t \int_{|z|<1} \gamma(X_i, z) \tilde{J}(ds, dz) + \int_{t_i}^t \int_{|z|\geq 1} \delta(X_i, z) J(ds, dz) - (t - t_i) \int_{|z|\geq 1} \delta(X_i, z) \nu(dz),$$

$$J^{**} = \int_{|z|<1} \gamma(X_i, z)^2 \nu(dz) + \int_{|z|\geq 1} \delta(X_i, z)^2 \nu(dz).$$

The compensator of the large-jump integral over  $[t_i, t]$  is  $(t - t_i) \int \delta \nu$ , proportional to the elapsed time—not  $h \int \delta \nu$ , as an earlier version had it; with the correct compensator,  $\mathbb{E}_i[J_t^*] = 0$  and  $\mathbb{E}_i[(J_t^*)^2] = (t - t_i) J^{**}$  (Lemma E.2). Higher conditional moments of  $J^*$  do *not* follow the Gaussian pattern: the third moment is  $\mathbb{E}_i[(J_t^*)^3] = (t - t_i) \int (\gamma^3 + \delta^3) d\nu$ , of order  $h$  rather than  $h^2$ , which is why the constant  $M_3$  appears in the theorem below.

**Theorem 5.3.** *Assume (L1) with  $M_3 < \infty$ , (L2) with  $m = \max(4, k + 1)$ ,  $f \in C_b^{k+1}$  jointly,  $\hat{g}, \hat{\gamma}(\cdot, z) \in C_b^{k+1}$ ,  $\gamma(\cdot, z), \delta(\cdot, z) \in C_b^2$  dominated by  $\rho_1, \rho_2$ , and  $\phi \in C^4$  with polynomially bounded derivatives. Then the Lévy  $k$ -ABM satisfies*

$$\text{Err}(T, h) \leq CT \left[ h(\varepsilon^2 + \mathbb{E}[J^{**}] + M_3) + h^k \left( \|\partial_x u\|_\infty \|F^{(k+1)}\|_\infty + \varepsilon^2 \|\partial_{xx} u\|_\infty \|G^{(k+1)}\|_\infty \right) \right] \\ + CT h^{3/2} \sqrt{\mathbb{E}[(J^{**})^3]},$$

where  $F, G$  are as in Lemma 4.9 and  $C$  depends on the norms in (L1)–(L2). In particular the scheme has weak order 1, with the  $h^k$  drift-extrapolation gain persisting in the small-noise, small-jump regime: if additionally  $\gamma = \varepsilon \hat{\gamma}$  and  $\delta \equiv 0$ , then  $\mathbb{E}[J^{**}] = O(\varepsilon^2)$ ,  $M_3 = O(\varepsilon^3)$ , and  $\text{Err}(T, h) = O(h\varepsilon^2 + h^k)$ .

*Proof.* See Appendix E. The structure follows Theorem 4.10: Itô’s formula for jump processes applied to  $u(t, \tilde{X}_t)$  produces, beyond the drift and diffusion mismatch terms of (7), jump mismatch terms  $e_i^\gamma$  and  $e_i^\delta$  comparing the generator’s jump integrals at the frozen coefficient  $\gamma(X_i, z)$  with those at  $\gamma(\tilde{X}_{t-}, z)$ . The naive estimate of these terms via  $\|\tilde{X}_{t-} - X_i\|_{L_2} = O(\sqrt{h})$  produces only  $O(\sqrt{h})$  per unit time, which is insufficient; Lemma E.3 improves this to  $O(h)$  per unit time by a second-order expansion in the frozen variable combined with a Dynkin/martingale pairing, using only  $\int \rho_1^2 d\nu < \infty$  for the small jumps. The third-moment structure of  $J^*$  enters through the Taylor remainder of  $f$  and contributes the  $M_3$  and  $h^{3/2}$  terms, the latter via Kunita’s inequality [18,

Theorem 4.4.23]. ■

## 6 Conclusion

We developed a general framework for the weak error analysis of stochastic linear multistep methods, based on a reduction of the scheme to a driven linear recurrence, conditional moment expansions, and the Kolmogorov backward equation. The framework yields an algorithmic weak error expansion for a broad class of SLMMs (Theorem 3.4), explicit first-order expansions for MAB2 and two-step Maruyama schemes (Theorems 4.2 and 4.6)—including a characterization of weak order 1 for the latter in which, unlike in the mean-square theory, only the square  $(\sum_j \gamma_j)^2$  of the noise weight sum is constrained—and small-noise error bounds for Adams–Bashforth-type schemes exhibiting a genuine asymptotic advantage over one-step methods (Theorem 4.10), together with an extension to Lévy-driven equations (Proposition 5.1, Theorem 5.3). Natural directions for future work include implicit and higher-weak-order multistep schemes, multidimensional and degenerate-noise settings, and sharp constants in the Lévy case.

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## Note on Generative AI Use

This manuscript was initially prepared without AI use. Later, it was put through Claude Fable 5, which made minor corrections and cleaned up/fleshed out some arguments. The ideas and main results are human-generated, but most of the writing in the appendix is due to Claude. A notable AI contribution was an error found in the original proof of Theorem 4.10, which Claude identified and fixed by introducing Lemma 4.9.



## A Auxiliary lemmas

Throughout the appendices,  $C$  denotes a constant depending only on  $T, k$ , the scheme coefficients, and the norms in (A3) (or (L1)–(L2)), whose value may change between occurrences. All schemes satisfy (A1)–(A4).

The first lemma provides uniform moment bounds for the derivative of the scheme with respect to a single Wiener increment; the second converts them into the pairing bound used repeatedly in the main text. For a smooth explicit scheme, the map  $\Delta W_j \mapsto X_m$  ( $m \geq j$ ) is differentiable for each fixed realization of the other increments; write  $D_j X_m := \partial X_m / \partial(\Delta W_j)$ .

**Lemma A.1.** *Assume the diffusion terms of (1) are of Maruyama type or MAB2 type (finitely many terms  $q(X, \cdot)$  times  $I_1$  or  $I_{1,1}$  with  $q \in C_b^1$ ). Then there is  $h_0 > 0$  such that for  $h \leq h_0$ ,*

$$\sup_{j \leq m \leq N} \mathbb{E}[(D_j X_m)^2] \leq C.$$

*Proof.* Differentiating the scheme recurrence in  $\Delta W_j$  gives, for  $m > j + k$ ,

$$\sum_{i=0}^k \alpha_i D_j X_{m-i} = h \sum_{i=1}^k \beta_i \partial_x f(X_{m-i}) D_j X_{m-i} + \sum_{i=1}^k \partial_x \Gamma_i(X_{m-i}) D_j X_{m-i} I^{t_{m-i}, t_{m-i+1}},$$

with an inhomogeneous injection at the steps  $m \in \{j, \dots, j+k\}$  where  $\Delta W_j$  appears explicitly (e.g.  $\gamma_1 g(X_{j-1})$  at  $m = j$ ); the injections are bounded in every  $L_p$  by (A3). By Lemma 3.1 applied to this linear recurrence with the homogeneous part  $\sum \alpha_i D_j X_{m-i} = 0$  moved to the left,

$$D_j X_m = (\text{injection terms}) + \sum_{i=j}^m G_{m-i} \left[ h b_i + \sigma_i I^{(i)} \right],$$

where  $b_i, \sigma_i$  are bounded multiples of  $D_j X_{i-1}, \dots, D_j X_{i-k}$  and  $I^{(i)}$  are the (mean-zero, variance- $O(h)$ ) stochastic factors. Let  $v_m = \mathbb{E}[(D_j X_m)^2]$ . Squaring and taking expectations, the drift part contributes, by Cauchy–Schwarz on the (boundedly weighted, since  $\sup_m \sum_i |G_{m-i}| h \leq C$ ) sum, at most  $Ch \sum_i G\text{-weights} \cdot \max_{i' \leq i} v_{i'}$ ; the martingale part contributes, by conditional orthogonality of the  $I^{(i)}$  across steps (condition on  $\mathcal{F}$  at the larger left endpoint) at most  $Ch \sum_i \theta^{2(m-i)} \max_{i' < i} v_{i'} \leq \frac{Ch}{1-\theta^2} \max_{i' < m} v_{i'}$ . Hence  $v_m \leq C_0 + C_1 h \sum_{i < m} w_{m-i} \bar{v}_i$  with summable weights and  $\bar{v}_i = \max_{i' \leq i} v_{i'}$ , and the discrete Grönwall inequality gives  $\bar{v}_N \leq C_0 e^{C_1 T}$ . ■

**Lemma A.2** (Pairing bound). *Let the scheme be as in Lemma A.1, and let*

$$\Phi_m = \Psi(X_m, X_{m-1}, \dots, X_{m-k}, t_m)$$

with  $\Psi \in C^1$  having polynomially bounded derivatives. Then, for all  $m \leq N$ ,

$$|\mathbb{E}[\Delta x_m \Phi_m]| \leq C_\Psi h.$$

More generally, if  $Y$  is any  $\mathcal{F}$ -adapted random variable of the form  $Y = q \Delta W_j$  or  $Y = q I_{1,1}^{t_{j-1}, t_j}$  with  $q$  bounded in  $L_4$  and  $\mathcal{F}_{t_{j-1}}$ -measurable, then  $|\mathbb{E}[Y \Phi_m]| \leq C_\Psi h$  for  $m \geq j$ .

*Proof.* By (4) and the geometric decay of the kernel,  $\Delta x_m$  is a sum of three parts: a transient satisfying  $|d_m - d_{m-1}| \leq C\theta^m \max_{i < k} |X_i - X_{i-1}| = O_{L_2}(\theta^m h^{1/2})$ , which pairs with bounded  $\Phi_m$  at order  $O(\theta^m h^{1/2})$ ; drift terms with weights  $hG^*$ , each contributing at most  $Ch\theta^{m-i}$  after Cauchy–Schwarz; and noise terms  $c g_{j-1} \Delta W_j$  or  $c q I_{1,1}^{(j)}$  with geometric weights  $|c| \leq C\theta^{m-j}$ . It therefore suffices to prove the second statement. For  $Y = q \Delta W_j$ : conditionally on all increments except  $\Delta W_j$ , apply Stein’s identity  $\mathbb{E}[\xi F(\xi)] = h \mathbb{E}[F'(\xi)]$  for  $\xi \sim N(0, h)$  to the  $C^1$  function  $\xi \mapsto q \Phi_m(\xi)$  (with  $q$  independent of  $\xi$ ):

$$\mathbb{E}[q \Delta W_j \Phi_m] = h \mathbb{E}\left[q \sum_{i=0}^k \partial_i \Psi \cdot D_j X_{m-i}\right],$$

which is bounded by  $C_\Psi h$  using Cauchy–Schwarz, Lemma A.1, and the moment bounds of (A3). For  $Y = q I_{1,1}$  with  $I_{1,1} = \frac{1}{2}(\Delta W_j^2 - h)$ : apply Stein’s identity twice (equivalently,  $\mathbb{E}[(\xi^2 - h)F(\xi)] = h^2 \mathbb{E}[F''(\xi)]$ ), yielding a bound  $C_\Psi h^2$  after using the second-derivative analogue of Lemma A.1, which holds by the same Grönwall argument. Summing the geometric weights over  $j \leq m$  multiplies the bound by  $\sum_j \theta^{m-j} \leq (1 - \theta)^{-1}$ . ■

**Lemma A.3** (Conditional moment expansion). *Under (A1)–(A3) with  $m = 2(S+2)$ , for every  $b \geq 1$  and  $S \geq 1$  there exist  $\mathcal{F}_{t_{n-1}}$ -measurable random variables  $C_{b,s,n}$ , given by explicit polynomials in  $h$ -free quantities (values of  $f$ , the  $\Gamma$ -coefficients and their derivatives at the nodes  $X_{n-1}, \dots, X_{n-k}$ , past increments  $\Delta x_{n-1}, \dots, \Delta x_{n-k+1}$ , and past single-interval iterated integrals), such that*

$$\mathbb{E}_{n-1}[\Delta x_n^b] = \sum_{2s \leq b}^{2s \leq 2S+1} h^{s/2} C_{b,s,n} + \varrho_{b,n}, \quad \mathbb{E}|\varrho_{b,n}| \leq Ch^{S+1},$$

where  $C_{b,s,n} = 0$  whenever  $b + s$  is odd in the pure-Maruyama case.

*Proof.* By (3),  $\Delta x_n$  splits as  $K_n$  plus the  $\Gamma$ -terms over  $[t_{n-1}, t_n]$ , where

$$K_n = - \sum_{i \geq 1} \gamma_i \Delta x_{n-i} + (\text{drift and past-interval diffusion terms of } e_n)$$

is  $\mathcal{F}_{t_{n-1}}$ -measurable. The new randomness enters only through iterated integrals over  $[t_{n-1}, t_n]$  with integrands evaluated along the path; Itô–Taylor expanding each integrand about  $(X_{n-1}, t_{n-1})$  to order  $S$  [2, Chapter 5] rewrites each such term as an adapted coefficient times a constant-integrand iterated integral over  $[t_{n-1}, t_n]$ , plus a remainder of  $L_2$  norm  $O(h^{(S+2)/2})$ . Expanding

$\Delta x_n^b$  by the multinomial theorem, the conditional expectation of each monomial factorizes: the  $\mathcal{F}_{t_{n-1}}$ -measurable part comes out, and the product of constant-integrand iterated integrals over the single interval has conditional expectation computable in closed form as a polynomial in  $h$  (Theorem 3.3 and the product formula for Hermite polynomials), homogeneous of the appropriate half-integer degree. Collecting powers of  $h^{1/2}$  gives the expansion; the moment bounds (A3) and (5) control the remainder. ■

**Lemma A.4** (Uniform first-order accuracy). *Let the scheme be MAB2, or a two-step Maruyama scheme satisfying the relations of Theorem 4.6(i). Let  $\psi \in C^4$  with polynomially bounded derivatives, and let  $n \leq N$ . Then*

$$|\mathbb{E}[\psi(X_n)] - \mathbb{E}[\psi(X(t_n))]| \leq C(1 + \|\psi\|_{C^4, \text{pol}})h,$$

with  $C$  independent of  $n$  and  $\psi$  except through the indicated norm.

*Proof.* Run the argument of Theorem 4.2 (resp. Appendix C) with horizon  $t_n$  and terminal function  $\psi$ : let  $u^\psi$  solve the KB-PDE on  $[0, t_n]$  with  $u^\psi(t_n, \cdot) = \psi$ ; by (A3) its spatial derivatives up to order 4 are polynomially bounded uniformly in  $n$ , with norms controlled by  $\|\psi\|_{C^4, \text{pol}}$  [10]. The local analysis of Steps 1–2 there shows each per-step error is  $O(h^2)$  (all that is used of the test function is four bounded-derivative orders), whence the total is  $Nh^2 \cdot C = O(h)$ . No marginal replacement (Step 3(ii)) is used in obtaining the  $O(h^2)$  per-step bound, so the argument is not circular. ■

## B Proof of Theorem 3.4

We prove the theorem by induction on the expansion order  $p \geq 0$ : we construct constants  $c_0, \dots, c_p$  and show  $\mathcal{E}(T, h) = \sum_{s \leq p} c_s h^{\gamma+s} + O(h^{\gamma+p+1})$ , for each  $p \leq r$ .

*Setup.* As in the main text,

$$\mathcal{E}(T, h) = \sum_{n=k}^N \mathbb{E}[u(t_n, X_{n-1} + \Delta x_n) - u(t_{n-1}, X_{n-1})] + \mathcal{B}_h,$$

where the boundary term  $\mathcal{B}_h$  collects the starter steps; by (A4) and the martingale property of  $u(t, X(t))$ ,  $\mathcal{B}_h = O(h^{\gamma+1}) + \sum_{s \leq p} b_s h^{\gamma+s}$  with computable starter-dependent constants  $b_s$  (obtained by applying the very expansion below to the  $O(k)$  starter steps; since there are boundedly many such steps, any per-step expansion to order  $h^{\gamma+p+1}$  suffices, and the induction below applies verbatim).

*Step 1: local expansion in computable form.* Fix a truncation order  $S = 2(p+2)$ . Taylor expanding (2) to total order  $S$  and substituting Lemma A.3,

$$\mathbb{E}[u(t_n, X_n) - u(t_{n-1}, X_{n-1})] = \sum_{s=2}^{2S+1} h^{s/2} \mathbb{E}[\mathcal{T}_{s,n}] + O(h^{p+\gamma+2}), \quad (8)$$

where each  $\mathcal{T}_{s,n}$  is an explicit finite sum of terms of the form

$$\tau = A_n \cdot \partial u(t_{n-1}, X_{n-1}),$$

with  $\partial u$  a spatial-temporal derivative of  $u$  of total order  $\leq S$  and  $A_n$  an  $\mathcal{F}_{t_{n-1}}$ -measurable polynomial in: node values of the coefficients and their derivatives, past increments  $\Delta x_{n-1}, \dots, \Delta x_{n-k+1}$ , and past single-interval iterated integrals with adapted coefficients. Only *even*  $s$  contribute at leading orders after the pairing step below; half-integer powers carry odd noise degree and will be shown to be expressible in integer orders.

*Step 2: reduction of adapted factors (the re-expansion).* Call a term  $\tau$  *reduced* if  $A_n$  is a function of  $(X_{n-1}, t_{n-1})$  alone. We claim: every term  $h^{s/2} \mathbb{E}[\tau]$  with  $\tau$  as above can be rewritten, up to  $O(h^{p+\gamma+2})$ , as a finite sum  $\sum_{s' \geq s} h^{s'/2} \mathbb{E}[\tau']$  of *reduced* terms with the same structure, where moreover only integer total powers  $h^{s'/2}$ ,  $s'$  even, survive. The proof of the claim is itself an induction on the *defect* of  $\tau$ , defined as the total noise degree of past-increment and past-integral factors in  $A_n$ . If the defect is 0, then  $A_n$  depends on  $X_{n-1}, \dots, X_{n-k}$  only through coefficient values; writing  $X_{n-i} = X_{n-1} - (\Delta x_{n-1} + \dots + \Delta x_{n-i+1})$  and Taylor expanding each coefficient about  $X_{n-1}$  converts  $\tau$  into reduced terms plus terms whose defect is positive but whose formal order is strictly larger; since each past increment carries  $L_2$  size  $O(h^{1/2})$  (Lemma 3.5), after finitely many steps the formal order exceeds  $p + \gamma + 2$  and the remainder is absorbed. If the defect is positive,  $A_n$  contains a factor which is a past iterated integral times an adapted coefficient, say  $q I^{(j)}$  over  $[t_{j-1}, t_j]$  with  $j \leq n-1$ . Apply the pairing identities of Lemma A.2 (Stein's identity, iterated as needed): this converts  $\mathbb{E}[\tau]$  into  $h^{\deg}$  times expectations of terms in which  $I^{(j)}$  has been replaced by derivatives  $D_j$  of the remaining factors; each application strictly reduces the noise degree while multiplying by an explicit power of  $h$ , and the derivative factors  $D_j X_m$  are themselves expandable: they satisfy the linear recurrence of Lemma A.1, whose kernel is the same geometric  $G^*$ , so that  $D_j X_m = (\text{explicit geometric weight}) \cdot (\text{coefficient values}) + (\text{higher-defect terms})$  by the same unrolling. Every replacement is explicit, and the geometric decay  $\theta^{n-j}$  of all memory weights makes the sums over the past index  $j$  absolutely convergent, uniformly in  $n$ ; summing the geometric series produces the closed-form constants (rational functions of the scheme coefficients, such as  $G_\infty = 1/\rho'(1)$ ) that appear in the final coefficients. Odd noise degrees have vanishing expectation at every stage, which eliminates half-integer powers. This proves the claim; note each step of the inner induction is a finite symbolic operation, which is the algorithmic content of the theorem.

*Step 3: cancellation and summation.* After Step 2, (8) reads

$$\mathbb{E}[u(t_n, X_n) - u(t_{n-1}, X_{n-1})] = \sum_{s=1}^{p+1} h^{\gamma+s} \mathbb{E}[\Upsilon_s(t_{n-1}, X_{n-1})] + O(h^{\gamma+p+2}),$$

where each  $\Upsilon_s$  is an explicit smooth functional of  $u$  and the coefficients, and where the terms of

order  $h^{\leq \gamma}$  have cancelled: the order- $h$  reduced content is  $h(\partial_t u + \bar{f}\partial_x u + \frac{1}{2}\bar{\sigma}^2\partial_{xx}u)$  with effective coefficients  $\bar{f}, \bar{\sigma}^2$  determined by the geometric sums, and weak consistency of order  $\gamma$  is precisely the statement that these match  $f$  and  $g^2$  so that the KB-PDE annihilates this bracket (Theorem 4.6 makes this matching explicit for two-step Maruyama schemes). Finally, for each  $s \leq p$ , replace  $\mathbb{E}[\Upsilon_s(t_{n-1}, X_{n-1})]$  by  $\mathbb{E}[\Upsilon_s(t_{n-1}, X(t_{n-1}))]$  at cost  $O(h^\gamma)$  per step by the induction hypothesis at order  $p-s$  applied with test function  $\Upsilon_s(t_{n-1}, \cdot)$  (for  $s = p$  use the crude order- $\gamma$  bound, i.e. the base case  $p = 0$ , which requires no replacement, exactly as in Lemma A.4); the total cost is  $h^{\gamma+s} \cdot N \cdot O(h^{\gamma+p-s}) \cdot h = O(h^{\gamma+p+1})$  after accounting for the prefactor, and lower-order replacement errors are re-expanded recursively into higher coefficients, which is the standard Talay–Tubaro bootstrap [10]. Riemann summation ( $s \mapsto \mathbb{E}[\Upsilon_s(s, X(s))]$  is Lipschitz by Itô’s formula and (A3)) converts  $h \sum_n \rightarrow \int_0^T$ , at cost one further power of  $h$  per coefficient, itself re-expanded recursively. Collecting orders yields  $c_s = \int_0^T \mathbb{E}[\Upsilon_{s+1}(t, X(t))] dt + b_s + (\text{recursive corrections})$ , all computable, completing the induction. ■

## C Proof of Theorem 4.6

Throughout, all  $u$ -derivatives are evaluated at  $(t_{n-1}, X_{n-1})$  unless stated otherwise;  $U_1 = \partial_x u$ ,  $U_2 = \partial_{xx}u$ , etc. We abbreviate  $\Sigma_\beta = \beta_1 + \beta_2$ ,  $\Sigma_\gamma = \gamma_1 + \gamma_2$ , and recall  $\zeta_2 = \alpha_2$ ,  $\rho'(1) = 1 - \zeta_2$ ,  $|\zeta_2| < 1$ .

### Sufficiency

Taylor expanding as in (2) to fourth spatial order,

$$\mathbb{E}[u(t_n, X_n) - u(t_{n-1}, X_{n-1})] = h \mathbb{E}[\partial_t u] + A_n + \frac{1}{2}B_n + \mathcal{R}_n, \quad (9)$$

where

$$A_n := \mathbb{E}[\Delta x_n U_1], \quad B_n := \mathbb{E}[\Delta x_n^2 U_2],$$

and  $\mathcal{R}_n$  collects  $\mathbb{E}[h\Delta x_n \partial_{tx}u]$ ,  $\frac{1}{2}h^2\mathbb{E}[\partial_{tt}u]$ ,  $\frac{1}{6}\mathbb{E}[\Delta x_n^3 U_3]$ ,  $\frac{1}{24}\mathbb{E}[\Delta x_n^4 U_4]$ , and the fifth-order Taylor remainder. We first record  $\mathcal{R}_n = O(h^2)$ : the  $h\partial_{tx}u$  term is  $h \cdot O(h)$  by Lemma A.2;  $\mathbb{E}[\Delta x_n^4 U_4] = O(h^2)$  by Lemma 3.5 and Cauchy–Schwarz; and  $\mathbb{E}[\Delta x_n^3 U_3] = O(h^2)$  because after unrolling  $\Delta x_n$  by Lemma 4.4 each cubic monomial has odd total noise degree, and one application of Stein’s identity (as in Lemma A.2) converts it to  $h$  times a quadratic-degree expectation of size  $O(h)$ .

*Recursions for  $A_n$  and  $B_n$ .* From (6),  $\Delta x_n = \zeta_2 \Delta x_{n-1} + e_n$ . For the  $A$ -recursion, shift the evaluation point of  $U_1$ :

$$\mathbb{E}[\Delta x_{n-1} U_1(t_{n-1}, X_{n-1})] = A_{n-1} + \mathbb{E}[\Delta x_{n-1} (U_2(t_{n-2}, X_{n-2}) \Delta x_{n-1} + h \partial_{tx}u + O(\Delta x_{n-1}^2 + h^2))],$$

where the quadratic shift term is  $B_{n-1}$ , the  $h\partial_{tx}u$  term is  $O(h^2)$  by Lemma A.2, and the cubic term is  $O(h^2)$  as above. For the forcing,  $\mathbb{E}[e_n U_1]$ : the  $\gamma_1 g_{n-1} \Delta W_n$  term vanishes ( $\Delta W_n$  independent of  $\mathcal{F}_{t_{n-1}}$ , mean zero); the drift term gives  $h \mathbb{E}[(\beta_1 f_{n-1} + \beta_2 f_{n-2}) U_1] = h \Sigma_\beta \mathbb{E}[f U_1] + O(h^2)$ , using  $f_{n-2} - f_{n-1} = O_{L_2}(h^{1/2})$  paired via Lemma A.2; and the  $\gamma_2 g_{n-2} \Delta W_{n-1}$  term pairs with the  $\Delta W_{n-1}$ -dependence of  $U_1(t_{n-1}, X_{n-1})$ : expanding  $X_{n-1} = X_{n-2} + \Delta x_{n-1}$  and  $\Delta x_{n-1} = \zeta_2 \Delta x_{n-2} + e_{n-1}$ , only the  $\gamma_1 g_{n-2} \Delta W_{n-1}$  constituent of  $e_{n-1}$  correlates at leading order, giving  $\gamma_1 \gamma_2 h \mathbb{E}[g^2 U_2] + O(h^2)$  (Stein's identity plus Lemma A.1 for the remainder). Altogether

$$A_n = \zeta_2 A_{n-1} + \zeta_2 B_{n-1} + h \Sigma_\beta \mathbb{E}[f U_1]_n + h \gamma_1 \gamma_2 \mathbb{E}[g^2 U_2]_n + O(h^2), \quad (10)$$

where the subscript  $n$  records that the smooth functionals are evaluated along the scheme at step  $n$  (their step-to-step variation is  $O(h)$ , by the same pairing argument, a fact used below). For the  $B$ -recursion, expand  $\Delta x_n^2 = \zeta_2^2 \Delta x_{n-1}^2 + 2\zeta_2 \Delta x_{n-1} e_n + e_n^2$ : the shift of  $U_2$  across one step costs  $\mathbb{E}[\Delta x_{n-1}^2 \cdot U_3 \Delta x_{n-1}] = O(h^2)$  (cubic, as above) plus  $O(h^2)$ ;  $\mathbb{E}[e_n^2 U_2] = h(\gamma_1^2 + \gamma_2^2) \mathbb{E}[g^2 U_2]_n + O(h^2)$ , since the  $\Delta W_n$ -square term gives  $\gamma_1^2 h$ , the  $\Delta W_{n-1}$ -square term gives  $\gamma_2^2 h$  after replacing  $\Delta W_{n-1}^2$  by  $h$  (the fluctuation  $\Delta W_{n-1}^2 - h$  pairs at order  $h^2$  by the second-order Stein identity), the cross-noise term has a free factor  $\Delta W_n$ , and drift crosses are  $O(h^{3/2})$ -degree odd terms pairing to  $O(h^2)$ ; and  $2\zeta_2 \mathbb{E}[\Delta x_{n-1} e_n U_2] = 2\zeta_2 \gamma_1 \gamma_2 h \mathbb{E}[g^2 U_2]_n + O(h^2)$ , the surviving pairing being between the  $\gamma_1 g_{n-2} \Delta W_{n-1}$  constituent of  $\Delta x_{n-1}$  and the  $\gamma_2 g_{n-2} \Delta W_{n-1}$  term of  $e_n$ ; the potentially dangerous term  $2\zeta_2^2 \gamma_2 \mathbb{E}[\Delta x_{n-2} g_{n-2} \Delta W_{n-1} U_2]$  has the form  $O(h) \cdot \mathbb{E}[\Delta x_{n-2} \cdot (\text{smooth adapted})] = h \cdot O(h)$  by Lemma A.2. Altogether

$$B_n = \zeta_2^2 B_{n-1} + h(\gamma_1^2 + \gamma_2^2 + 2\zeta_2 \gamma_1 \gamma_2) \mathbb{E}[g^2 U_2]_n + O(h^2). \quad (11)$$

*Fixed points and transients.* Consider first (11). Write it as  $B_n = \zeta_2^2 B_{n-1} + F_n$  with  $F_n = O(h)$  and  $|F_n - F_{n-1}| = O(h^2)$  (step-to-step variation of the smooth functional). Iterating,  $B_n = \sum_{m \leq n} \zeta_2^{2(n-m)} F_m + \zeta_2^{2(n-n_0)} B_{n_0}$ ; splitting  $F_m = F_n + (F_m - F_n)$  with  $|F_m - F_n| \leq (n-m) O(h^2)$  and  $\sum_m \zeta_2^{2(n-m)} (n-m) \leq C$ ,

$$B_n = \frac{hV}{1} \mathbb{E}[g^2 U_2]_n + O(h^2) + O(\theta^n) \cdot O(h), \quad V := \frac{\gamma_1^2 + \gamma_2^2 + 2\zeta_2 \gamma_1 \gamma_2}{1 - \zeta_2^2},$$

where the transient uses  $B_{n_0} = O(h)$  (Lemma A.2 and Cauchy-Schwarz give  $B_m = O(h)$  for every  $m$ ) and  $\theta = \zeta_2^2$ . The same treatment of (10), with forcing  $\zeta_2 B_{n-1} + O(h)$ -smooth terms, gives

$$A_n = \frac{1}{1 - \zeta_2} \left( \zeta_2 B_n + h \Sigma_\beta \mathbb{E}[f U_1]_n + h \gamma_1 \gamma_2 \mathbb{E}[g^2 U_2]_n \right) + O(h^2) + O(\theta^n) O(h).$$

Substituting into  $A_n + \frac{1}{2} B_n$  and simplifying the  $g^2 U_2$  coefficient over the common denominator

$2(1 - \zeta_2)^2$ :

$$\frac{\gamma_1^2 + \gamma_2^2 + 2\zeta_2\gamma_1\gamma_2}{1 - \zeta_2^2} \cdot \frac{1 + \zeta_2}{2(1 - \zeta_2)} + \frac{\gamma_1\gamma_2}{1 - \zeta_2} = \frac{\gamma_1^2 + \gamma_2^2 + 2\gamma_1\gamma_2}{2(1 - \zeta_2)^2} = \frac{1}{2} \left( \frac{\Sigma_\gamma}{\rho'(1)} \right)^2.$$

Hence, with  $w_\beta := \Sigma_\beta/\rho'(1)$  and  $w_\gamma := \Sigma_\gamma/\rho'(1)$ ,

$$(9) = h \mathbb{E} \left[ \partial_t u + w_\beta f \partial_x u + \frac{1}{2} w_\gamma^2 g^2 \partial_{xx} u \right]_n + O(h^2) + O(\theta^n) O(h). \quad (12)$$

If  $w_\beta = 1$  and  $w_\gamma^2 = 1$ , the bracket is  $(\partial_t + \mathcal{L})u = 0$  pointwise by the KB-PDE, so each step contributes  $O(h^2) + O(\theta^n)O(h)$ ; summing over  $n \leq N = T/h$  and adding the  $O(h^2)$  starter term gives  $\mathcal{E}(T, h) = O(h)$ , i.e. weak order 1. (With more bookkeeping, exactly as in Theorem 4.2, the  $O(h^2)$  terms can be organized into an explicit  $c_0 h + O(h^2)$  expansion; weak order 1 requires only the bound.)

### Necessity

Assume weak order 1 for all data as in (A3). If  $w_\beta \neq 1$  or  $w_\gamma^2 \neq 1$ , then by (12)

$$\begin{aligned} \mathcal{E}(T, h) &= h \sum_n \mathbb{E} \left[ (w_\beta - 1) f \partial_x u + \frac{1}{2} (w_\gamma^2 - 1) g^2 \partial_{xx} u \right]_n + O(h) \\ &\longrightarrow \int_0^T \mathbb{E} \left[ (w_\beta - 1) f \partial_x u + \frac{1}{2} (w_\gamma^2 - 1) g^2 \partial_{xx} u \right] dt, \end{aligned}$$

so it suffices to exhibit data making the limit nonzero. (The marginal replacement inside the integral uses Lemma A.4-type continuity of the functional in  $h$ , or one may simply argue directly on the two explicit examples, as we now do, without invoking (12) at all.)

*Drift relation.* Take  $f \equiv 1$ ,  $g \equiv 0$ ,  $\phi(x) = x$ ,  $X_0 = X_1 = 0$ . The scheme is the deterministic recurrence  $X_n = -\alpha_1 X_{n-1} - \alpha_2 X_{n-2} + h \Sigma_\beta$ . Trying  $X_n = cnh$ :  $cnh(1 + \alpha_1 + \alpha_2) - ch(\alpha_1 + 2\alpha_2) = h \Sigma_\beta$ ; the first bracket vanishes by  $\rho(1) = 0$  and  $-(\alpha_1 + 2\alpha_2) = \rho'(1)$ , so the particular solution is  $X_n = \frac{\Sigma_\beta}{\rho'(1)} nh$  and the general solution adds  $a + b\zeta_2^n$  with  $a, b$  fixed by the initial data, uniformly bounded. Hence  $\mathbb{E}[\phi(X_N)] - \mathbb{E}[\phi(X(T))] = (w_\beta - 1)T + O(1) \cdot (h + \theta^N)$ , which tends to the nonzero constant  $(w_\beta - 1)T$  unless  $w_\beta = 1$ . So weak order 1 (indeed, weak convergence) forces  $\Sigma_\beta = \rho'(1)$ .

*Noise relation.* Take  $f \equiv 0$ ,  $g \equiv 1$ ,  $\phi(x) = x^2$ ,  $X_0 = X_1 = 0$ . Then  $X(T) = W_T$  and  $\mathbb{E}[\phi(X(T))] = T$ . By Lemma 4.4 with  $g \equiv 1$  and  $d_n \equiv 0$ ,

$$X_N = \sum_{j=1}^N (\gamma_1 G_{N-j} + \gamma_2 G_{N-j-1}) \Delta W_j, \quad \mathbb{E}[X_N^2] = h \sum_{j=1}^N (\gamma_1 G_{N-j} + \gamma_2 G_{N-j-1})^2.$$

Since  $G_m \rightarrow G_\infty = 1/\rho'(1)$  geometrically, all but  $O(\log(1/h))$  of the  $N$  summands equal  $(\Sigma_\gamma G_\infty)^2 + O(\theta^{N-j})$ , whence  $\mathbb{E}[X_N^2] = T w_\gamma^2 + O(h \log(1/h))$ . Weak order 1 forces  $w_\gamma^2 = 1$ , i.e.  $\Sigma_\gamma^2 = \rho'(1)^2$ . ■

## D Proofs of Lemmas 4.8 and 4.9

*Proof of Lemma 4.8.* For (a): for  $t \in [t_n, t_{n+1}]$  and  $1 \leq j \leq k$ ,  $|t - t_{n-j}| \leq (j+1)h$ , so the product is at most  $\prod_{j=1}^k (j+1)h = (k+1)!h^k$ . For (b): the numerator of  $\ell_j$  is bounded by  $\prod_{i \neq j} (i+1)h = \frac{(k+1)!}{j+1}h^{k-1}$  and its denominator equals  $\prod_{i \neq j} |j-i|h = (j-1)!(k-j)!h^{k-1}$ , so  $|\ell_j(t)| \leq \frac{(k+1)!}{(j+1)(j-1)!(k-j)!} \leq (k+1)!$ , and summing over  $j$  gives  $C_k \leq k(k+1)!$ . (Sharper constants are available but not needed.) For (c): the Lagrange remainder formula gives, for each  $t$  in the stated interval and some  $\xi_t$  in the node hull,  $\psi(t) - L_k\psi(t) = \frac{\psi^{(k+1)}(\xi_t)}{k!} \prod_{j=1}^k (t - t_{n-j}) \cdot \frac{k!}{k!}$ ; more precisely the standard formula has denominator  $k!$  for  $k$  nodes, and combining with (a) yields the bound  $\|\psi^{(k+1)}\|_\infty \frac{(k+1)!}{k!} h^k \cdot \frac{1}{k+1} \leq \|\psi^{(k+1)}\|_\infty h^k$ . ■

*Proof of Lemma 4.9.* (a) Let  $Y_t = X(t) - x_t^0$ . Then  $dY = (f(X) - f(x^0))dt + \varepsilon \hat{g}(X) dW$ , so by boundedness of  $\partial_x f$ , Itô isometry, and Grönwall,  $\sup_{t \leq T} \mathbb{E}[Y_t^2] \leq C\varepsilon^2$ ; the same argument at fourth moments gives  $\sup_t \|Y_t\|_{L_4} \leq C\varepsilon$ . Now let  $Z_t = X(t) - x_t^0 - \varepsilon R_t$ . Then

$$dZ = (f(X) - f(x^0) - \varepsilon \partial_x f(x^0)R)dt + \varepsilon(\hat{g}(X) - \hat{g}(x^0))dW.$$

The drift is  $\partial_x f(x^0)Z + O(|Y|^2)$  pointwise by Taylor's theorem with  $\|\partial_{xx} f\|_\infty$ , and the diffusion coefficient is bounded by  $\varepsilon \|\partial_x \hat{g}\|_\infty |Y|$ . Hence  $\mathbb{E}[Z_t^2] \leq C \int_0^t (\mathbb{E}[Z^2] + \mathbb{E}[Y^4] + \varepsilon^2 \mathbb{E}[Y^2])ds \leq C \int_0^t \mathbb{E}[Z^2]ds + C\varepsilon^4 t$ , and Grönwall gives (a).

(b) Let  $x_n^{0,h}$  be the  $k$ -step Adams–Bashforth iterates for  $\dot{x} = f(x, t)$  from exact starting values; the classical convergence theorem for zero-stable, order- $k$ -consistent linear multistep methods [16, Chapter III.4] gives  $\max_n |x_n^{0,h} - x_{t_n}^0| \leq Ch^k$ . Define  $R^h$  by the linearized scheme:

$$R_{n+1}^h = R_n^h + \int_{[t_n, t_{n+1}]} L_k[\partial_x f(x^{0,h})R^h](t) dt + \int_{[t_n, t_{n+1}]} L_k[\hat{g}(x^{0,h})](t) dW_t, \quad R_j^h = 0, \quad j < k,$$

where  $L_k$  interpolates the bracketed node values. Since the deterministic weights are bounded and the stochastic forcing is a mean-zero Itô integral with  $\mathcal{F}$ -adapted integrand, an induction gives  $\mathbb{E}[R_n^h] = 0$ ; moment bounds  $\max_n \|R_n^h\|_{L_4} \leq C$  follow from discrete Grönwall exactly as in Lemma A.1 (the scheme is one-step in the pair  $(X_n, \dots, X_{n-k+1})$  with bounded coefficients). Finally,  $Z_n^h := X_n - x_n^{0,h} - \varepsilon R_n^h$  satisfies a driven recurrence whose forcing per step is, by Taylor expansion of the interpolated node values, bounded in  $L_2$  by  $Ch(\max_{m \leq n} \|Z_m^h\|_{L_2} + \varepsilon^2 + h^k) + C\varepsilon\sqrt{h}(\max_m \|Z_m^h\|_{L_2} + \varepsilon + h^k)$ , the  $\sqrt{h}$  factor coming from the Itô integral of the diffusion mismatch; squaring, using orthogonality of the martingale increments, and applying discrete Grönwall yields  $\max_n \|Z_n^h\|_{L_2} \leq C(\varepsilon^2 + \varepsilon h^k)$ . (c) is immediate from (b) and Taylor expansion of  $f$  and  $\hat{g}^2$  about  $x_n^{0,h}$ , absorbing  $|x_n^{0,h} - x_{t_n}^0| \leq Ch^k$  into  $\eta$ . ■



## E Proofs for Section 5

We first record the integration-by-parts tool replacing Stein's identity, and the moment structure of  $J^*$ .

**Lemma E.1** (Mecke formula; see [18, Section 4.2]). *Let  $N$  be a Poisson random measure on  $[0, T] \times \mathbb{R}$  with intensity  $dt \otimes \nu$ , and let  $H(s, z, \omega)$  be measurable and integrable. Then*

$$\mathbb{E} \left[ \int H(s, z) N(ds, dz) \right] = \int \mathbb{E}[H(s, z) \circ \theta_{s,z}] ds \nu(dz),$$

where  $\theta_{s,z}$  adds an atom at  $(s, z)$  to the configuration. Consequently, for the compensated measure,  $\mathbb{E}[\int H \tilde{N}] = \int \mathbb{E}[H \circ \theta_{s,z} - H] ds \nu(dz)$  whenever both sides are defined.

**Lemma E.2.** *With  $J_t^*$ ,  $J^{**}$  as in Section 5 and (L1) with  $M_3 < \infty$ : conditionally on  $\mathcal{F}_{t_i}$ , for  $t \in [t_i, t_{i+1}]$ ,*

$$\mathbb{E}_i[J_t^*] = 0, \quad \mathbb{E}_i[(J_t^*)^2] = (t - t_i) J^{**}, \quad \mathbb{E}_i[(J_t^*)^3] = (t - t_i) \int (\gamma(X_i, z)^3 + \delta(X_i, z)^3) \nu(dz),$$

and  $\mathbb{E}_i[|J_t^*|^3] \leq C((t - t_i)M_3 + (t - t_i)^{3/2}(J^{**})^{3/2})$  (Kunita's inequality, [18, Theorem 4.4.23]).

*Proof.* Given  $\mathcal{F}_{t_i}$ ,  $J^*$  is a compensated Poisson stochastic integral with deterministic (frozen) integrand, hence a martingale in  $t$  with  $\mathbb{E}_i[J_t^*] = 0$ ; the second moment is the standard isometry for compensated Poisson integrals; the third moment follows from Itô's formula for  $(J^*)^3$ : the jump compensator term contributes  $\int ((J^* + \kappa)^3 - (J^*)^3 - 3(J^*)^2 \kappa) \nu = \int (3J^* \kappa^2 + \kappa^3) \nu$  with  $\kappa \in \{\gamma, \delta\}$ , and  $\mathbb{E}_i[J_s^*] = 0$  kills the middle term upon integration. The absolute bound is Kunita's inequality. ■

The key improvement over the naive  $O(\sqrt{h})$  bound for the jump mismatch is the following.

**Lemma E.3.** *Assume (L1)–(L2) with  $m \geq 4$  and  $\gamma(\cdot, z) \in C_b^2$  dominated by  $\rho_1$ . For  $t \in [t_i, t_{i+1}]$  define*

$$\psi_z(t, x, y) := u(t, x + \gamma(y, z)) - u(t, x) - \gamma(y, z) \partial_x u(t, x),$$

and  $e_i^\gamma := \mathbb{E} \int_{[t_i, t_{i+1}]} \int_{|z| < 1} [\psi_z(t, \tilde{X}_{t-}, \tilde{X}_{t-}) - \psi_z(t, \tilde{X}_{t-}, X_i)] \nu(dz) dt$ . Then  $|e_i^\gamma| \leq Ch^2$ , with  $C$  depending only on  $\int \rho_1^2 d\nu$ , the  $C_b^2$  norms in (L2),  $\|u\|_{C^{1,4}, \text{pol}}$ , and the moment bounds. The analogous quantity  $e_i^\delta$  built from  $\psi_z(t, x, y) = u(t, x + \delta(y, z)) - u(t, x)$  on  $\{|z| \geq 1\}$  satisfies the same bound with constant depending on  $\int_{|z| \geq 1} \rho_2 d\nu < \infty$ .

*Proof.* Fix  $t, z$  and expand in the third argument about  $y = X_i$  with integral remainder:

$$\psi_z(t, x, \tilde{X}_{t-}) - \psi_z(t, x, X_i) = \partial_y \psi_z(t, x, X_i) \Delta_t + \int_0^1 (1 - \lambda) \partial_{yy} \psi_z(t, x, X_i + \lambda \Delta_t) d\lambda \Delta_t^2,$$

with  $\Delta_t := \tilde{X}_{t-} - X_i$  and  $x = \tilde{X}_{t-}$ . The domination structure gives the crucial quadratic bounds

$$|\partial_y \psi_z| \leq |\partial_y \gamma| \cdot |\partial_x u(t, x + \gamma) - \partial_x u(t, x)| \leq \rho_1(z)^2 \|\partial_{xx} u\|, \quad |\partial_{yy} \psi_z| \leq C \rho_1(z)^2 \|u\|_{C^2},$$

and similarly each additional  $x$ -derivative of  $\partial_y \psi_z$  preserves the difference structure and hence the  $\rho_1^2$  domination, up to the fourth  $u$ -derivative. (Polynomially bounded rather than bounded  $u$ -derivatives are handled by the uniform moment bounds; we write sup norms for readability.)

*Quadratic remainder:*  $\mathbb{E}_i[\Delta_t^2] \leq C(t - t_i)(C_k^2 \varepsilon^2 + J^{**} + h) \leq Ch$ , so this term contributes at most  $C \rho_1(z)^2 h$  per  $(t, z)$ , hence  $Ch^2 \int \rho_1^2 d\nu$  after integrating  $dt \nu(dz)$ .

*Linear term:* write  $\Delta_t = D_t + M_t$ , where  $D_t = \int_{t_i}^t (P_k - \int_{|z| \geq 1} \delta \nu) ds$  is the finite-variation part and  $M_t = \int Q_k dW + J_t^*$  the martingale part. The drift part obeys  $|\mathbb{E}[\partial_y \psi_z \cdot D_t]| \leq C \rho_1^2 h$  per  $(t, z)$ , contributing  $Ch^2$  as before. For the martingale part, set  $\Phi_z(s) := \partial_y \psi_z(t, \tilde{X}_s, X_i)$  and apply Itô's product formula to  $s \mapsto \Phi_z(s^-) M_s$  on  $[t_i, t]$ , taking  $\mathbb{E}_i$ : the terms  $\int \Phi_z dM$  and  $\int M_{s-} d(\text{martingale part of } \Phi_z)$  vanish in expectation, leaving

$$\mathbb{E}_i[\Phi_z(t^-) M_t] = \mathbb{E}_i \int_{t_i}^t \left[ M_{s-} (\mathcal{A}_s \Phi_z)(s) + \frac{d\langle \Phi_z, M \rangle_s}{ds} \right] ds,$$

where  $\mathcal{A}_s$  is the (time-dependent, frozen-coefficient) generator of  $\tilde{X}$  acting on the  $x$ -argument of  $\partial_y \psi_z$ , and the bracket has density  $\partial_x \Phi_z \cdot Q_k(s)^2 + \int [\Phi_z(x + \text{jump}) - \Phi_z(x)] \cdot (\text{jump of } M) \nu$ . By the  $\rho_1^2$ -domination of  $\partial_y \psi_z$  and its  $x$ -derivatives up to second order (which is what  $\mathcal{A}_s$  requires; total  $u$ -regularity used: four spatial derivatives),  $\|\mathcal{A}_s \Phi_z\| \leq C \rho_1(z)^2$  and the bracket density is bounded by  $C \rho_1(z)^2 (\varepsilon^2 C_k^2 + J_{\text{abs}}^{**})$  with  $J_{\text{abs}}^{**} = \int \rho_1^2 d\nu + \int \rho_2^2 d\nu$ -type constants. Since  $\mathbb{E}_i[|M_s|] \leq (\mathbb{E}_i[M_s^2])^{1/2} \leq C\sqrt{h}$ , each  $(t, z)$  contributes at most  $C \rho_1(z)^2 h(\sqrt{h} + 1) \leq C \rho_1(z)^2 h$ , and integrating over  $t \in [t_i, t_{i+1}]$  and  $z$  gives  $|e_i^\gamma| \leq Ch^2$ . For  $e_i^\delta$ , the same argument runs with  $|\partial_y \psi_z| \leq \rho_2(z) \|\partial_x u\|$  (linear in  $\rho_2$  suffices, as  $\psi$  lacks the compensator subtraction), and  $\int_{|z| \geq 1} \rho_2 d\nu < \infty$  by Cauchy-Schwarz. ■

*Proof of Theorem 5.3.* Apply Itô's formula for jump diffusions to  $u(t, \tilde{X}_t)$  on  $[t_i, t_{i+1}]$  and take expectations; the KB equation  $\partial_t u + \mathcal{L}u = 0$  at  $(t, \tilde{X}_{t-})$  leaves the mismatch

$$e_i = \mathbb{E} \int_{[t_i, t_{i+1}]} \left[ \partial_x u(P_k - f(\tilde{X}_{t-})) + \frac{1}{2} \partial_{xx} u(Q_k^2 - g^2(\tilde{X}_{t-})) \right] dt - e_i^\gamma - e_i^\delta - e_i^c,$$

where  $e_i^\gamma, e_i^\delta$  are as in Lemma E.3 and  $e_i^c$  collects the compensator-drift mismatch

$$\partial_x u \cdot \left( \int_{|z| \geq 1} \delta(\tilde{X}_{t-}, z) \nu(dz) - \int_{|z| \geq 1} \delta(X_i, z) \nu(dz) \right),$$

which is bounded by the same Dynkin pairing as the drift part of Lemma E.3 by  $Ch^2$  per step. Lemma E.3 bounds  $e_i^\gamma + e_i^\delta$  by  $Ch^2$  per step, i.e.  $Ch$  in total: this is the  $h \mathbb{E}[J^{**}]$ -type contribu-

tion displayed (tracking constants, the  $\gamma$ -part carries  $\int \rho_1^2 d\nu$  and the quadratic remainder carries  $\mathbb{E}_i[\Delta_t^2] \ni (t - t_i)J^{**}$ ).

The drift and diffusion mismatches are decomposed as in Theorem 4.10 into fluctuation and interpolation parts. Two changes arise from the jumps. First, the conditional fluctuation about the conditional mean is now  $M_t = \int Q_k dW + J_t^*$ , no longer Gaussian: in the Taylor expansion of  $f$  about the conditional mean, the quadratic term contributes  $\frac{1}{2}\partial_{xx}f \cdot (\int_{t_i}^t Q_k^2 ds + (t - t_i)J^{**})$  (conditional independence of the Brownian and Poisson parts kills the cross term), of size  $Ch(\varepsilon^2 C_k^2 + J^{**})$  per  $t$ , hence  $Ch^2$  per step and  $Ch$  total, as displayed; the cubic term is controlled by Lemma E.2:  $\mathbb{E}_i[|M_t|^3] \leq C(hM_3 + h^{3/2}(J^{**})^{3/2} + h^{3/2}\varepsilon^3)$ , contributing  $h(hM_3 + h^{3/2}\mathbb{E}[(J^{**})^{3/2}])$  per step, i.e. the  $hM_3$  and  $h^{3/2}$  terms of the statement after summation (Jensen bounds  $\mathbb{E}[(J^{**})^{3/2}] \leq \sqrt{\mathbb{E}[(J^{**})^3]}$ ). Second, in the small-noise decomposition underlying the interpolation part, the residual process acquires a jump forcing; since the jump coefficients are frozen at  $X_i$  and enter the scheme additively with zero conditional mean (after compensation), the mean-zero structure of Lemma 4.9(b) persists, with  $L_2$  bounds now involving  $\mathbb{E}[J^{**}]$ ; the interpolation estimates (IIa)–(IIc) of Theorem 4.10 then apply verbatim, yielding the  $h^k$  terms displayed. Summing over the  $T/h$  steps completes the proof. ■

*Proof of Proposition 5.1.* The proof follows Appendix B line by line; we indicate the substitutions. *Regularity input:* the solution  $u$  of the integro-differential KB equation has  $m$  polynomially bounded spatial derivatives under (L1)–(L2); this is proved by Protter–Talay [17, Section 4] via the stochastic flow, and in [18, Chapter 6]. *Moment input:* the scheme’s iterates have uniformly bounded moments up to the order needed, by discrete Grönwall using (L1) and the isometry/Kunita bounds for compensated Poisson integrals [18, Theorem 4.4.23], exactly as in Lemma A.1. *Local expansion (Step 1):* the conditional moment expansion of Lemma A.3 extends: the new-interval randomness now includes  $\int \int \kappa(X_{n-1}, z) \tilde{N}$ -type integrals with frozen coefficients, whose conditional mixed moments with the Gaussian integrals factor by independence and are polynomials in  $(t - t_{n-1})$  and the  $\nu$ -moments of the coefficients (Lemma E.2 and its higher-order analogues, available under (L1)–(L2) with  $m(r)$  moments of  $\rho_1, \rho_2$ ; this is where  $m(r)$  enters). *Pairing (Step 2):* Stein’s identity is replaced by the Mecke formula (Lemma E.1): for an adapted factor times a past compensated jump integral,  $\mathbb{E}[\int H \tilde{N} \cdot \Phi] = \int \mathbb{E}[(H\Phi) \circ \theta_{s,z} - H\Phi] ds \nu(dz)$ , and the add-one-jump difference of the smooth functional  $\Phi$  is bounded by the jump size times  $D$ -type derivative bounds along the scheme, dominated by  $\rho_1(z)$  (resp.  $\rho_2$ ); the geometric memory decay from the strict root condition is unchanged, since it is a property of the deterministic kernel  $G^*$ . *Cancellation and summation (Step 3):* identical, with  $\mathcal{L}$  the integro-differential generator; the effective drift, diffusion, and jump intensities produced by the geometric sums must match those of  $\mathcal{L}$ , which is the weak consistency hypothesis of order  $\gamma$ . The bootstrap and Riemann summation are unchanged. ■

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